# Auditing Frozen Foundation-Model Embeddings: A Leakage-Controlled, Calibration-Aware Evaluation Protocol Stress-Tested on Single-Cell Genomics

Zeyu Fu

State Key Laboratory of Trauma and Chemical Poisoning, Institute of Combined Injury, Chongqing Engineering Research Center for Nanomedicine, College of Preventive Medicine, Army Medical University, Chongqing, China

fuzeyu99@126.com

2026-06-23

**Abstract**

Evaluating frozen foundation-model (FM) embeddings is unreliable when the scoring metrics are entangled with the methods they rank, when tokenizer coverage is confounded with representation quality, and when the exchangeability that underwrites uncertainty guarantees is broken by distribution shift. We present a reusable, deterministic evaluation protocol for pretrained representations under distribution shift, built on four domain-general moves: every agreement metric is paired with a non-linear ($k$NN) probe and a reference-free structure metric to expose label circularity; tokenizer/vocabulary coverage is treated as a measured variable rather than a fixed assumption; parity is argued with an equivalence test (TOST) rather than an absent $p$-value; and reliability is audited with split-conformal coverage, ECE calibration, and selective abstention. The protocol's signature move replaces binary leaderboard verdicts with a dose–response analysis that places the criticized method class on the curve, converting a rank into a manipulable, causal mechanism. We stress-test it on single-cell genomics—a demanding regime of batch/donor shift, circular annotation, and cross-species tokenizer mismatch—across four modalities, up to 24 leakage-controlled atlases, and five FM families. Three findings follow: zero-shot single-cell FMs neither beat nor trail simple baselines on matched vocabularies; the much-warned cross-batch calibration collapse is a general exchangeability failure of the data regime, not an FM defect; and which side "wins" is set by task and evaluation metric, not model scale or architecture. All findings are scoped to the zero-shot, frozen-embedding regime on coarse human cell types; fine-tuning and few-shot transfer are untested and can reverse them.

**Keywords.** representation evaluation; frozen foundation-model embeddings; distribution shift; evaluation-metric circularity; conformal prediction; calibration; equivalence testing; dose–response analysis; single-cell genomics.

## 1 Introduction

Foundation models [1, 2] are increasingly deployed as *frozen* feature extractors: a network pretrained on large unlabelled corpora is used, without task-specific training, to embed new inputs, and those embeddings are then ranked on leaderboards against cheaper baselines. How to evaluate such representations fairly is a general and unsettled question for neural learning systems: leaderboards compare a frozen model against baselines using agreement metrics that presuppose what counts as ground truth, and rarely audit whether the comparison is confounded or whether the resulting confidence is reliable under shift. Single-cell genomics is a sharp instance and a useful stress-test domain—models pretrained on millions of cells [3, 4] (scGPT [5], Geneformer [6], scFoundation [7], UCE [8], CellPLM [9], and spatial/chromatin variants [10–12]) are promised to recover cell types, integrate batches, and support downstream inference better than long-established, largely linear pipelines. A growing critical literature reports the opposite in the *zero-shot* setting: frozen embeddings do not outperform baselines (PCA, scVI [13], Harmony [14], HVG, logistic regression; additive/linear models for perturbation) [15], and a simple supervised linear probe can be competitive with them. But each result is typically one paper on one task, and the reliability axis—calibration, conformal coverage, selective abstention—is under-audited; to our knowledge no calibration or conformal audit of any scATAC foundation model exists. We scope every claim to the zero-shot, frozen-embedding regime; fine-tuned scFMs can reverse these results and are not extrapolated to.

The disputes on both sides share a methodological flaw that is not specific to genomics: the yardsticks are entangled with the methods they rank. When a benchmark's "ground truth" is itself derived in the representation space it is grading, agreement scores reward proximity to the label-generating process rather than to the underlying signal—label circularity, the representation-evaluation analogue of testing on a training signal. In single-cell data this is explicit: cell-type labels are produced by clustering and marker scoring [16] in the same expression space a linear baseline such as PCA occupies, so ARI, NMI, scIB, and label-AUROC can reward a method for recovering the clustering that defined the labels. A second confound is the tokenizer: a model with a fixed vocabulary silently drops unmappable inputs (gene symbols vs. Ensembl IDs, cross-species orthologs), so a benchmark may score a model's *coverage* while appearing to score its representation. A third bites at deployment: valid uncertainty sets from split-conformal prediction hold only under exchangeability [17], which the batch and donor shift pervading single-cell data routinely violates—and whether frozen embeddings repair, preserve, or worsen that failure has not been measured for any modality. Label circularity, tokenizer coverage, and exchangeability violation are general hazards of evaluating pretrained representations; genomics simply makes all three severe at once.

We therefore present the evaluation protocol as the contribution, and genomics as the stress-test domain that exposes what it catches. The protocol is a deterministic, leakage-controlled fair-evaluation and reliability audit for frozen embeddings under distribution shift. It pairs every headline agreement metric with a non-linear ($k$NN) probe and a reference-free, label-independent structure metric to separate representational quality from label circularity; it treats vocabulary coverage as a measured variable rather than an assumption; it argues parity with an equivalence test (TOST) instead of an absent $p$-value; and it audits reliability with split-conformal coverage, ECE calibration, and selective abstention. Its signature move replaces a binary leaderboard verdict with a graded *dose–response* in which the criticized method class is placed on the curve, turning a rank into a manipulable, causal mechanism whose dose is vocabulary coverage, batch-shift strength, or spatial-aggregation depth.

Applied to genomics, the protocol yields three provisional findings, each a demonstration of a distinct failure mode. First, on vocabulary-matched atlases frozen embeddings show no advantage over simple baselines—and no deficit: no FM family differs significantly from PCA under the non-linear probe (a boundary equivalence read as "no significant difference" rather than a clean parity pass), and the apparent baseline "win" dissolves into two artifacts the deconfounding tools isolate—a label circularity and a causal tokenizer-coverage threshold. Second, the much-warned cross-batch calibration collapse is a *general* exchangeability failure of the data regime, not an FM defect: it afflicts classical baselines as much as frozen embeddings, and is absent in scATAC.

Third, which method "wins" is governed by task and evaluation metric, not by model scale or architecture. We also report, to our knowledge, the first calibration/reliability audit of a single-cell ATAC modality (ChromFound and Atacformer) and the first head-to-head of zero-shot spatial FMs against a parameter-free niche baseline. All findings are scoped to the zero-shot, frozen-embedding regime on coarse human cell types; fine-tuning and few-shot transfer are untested and can reverse them. These are the results of a single practitioner on tractable subsets and may not survive at scale or under different recipes; Figure 1 orients the study design (A) and the reasoning route from the field's claim to the fairer test (B).

**Contributions.** The paper's deliverable is an evaluation instrument, not a verdict on foundation models.

- **A fair, leakage-controlled, reliability-aware evaluation protocol for frozen FM embeddings under distribution shift**—one deterministic, openly released pipeline that scores pretrained embeddings and their baselines on identical splits, pairing a representation axis (non-linear probe, reference-free structure metric, equivalence testing) with a reliability axis (conformal coverage, calibration, selective abstention).
- **Dose–response in place of binary leaderboard verdicts**—a graded curve on which the criticized method class is placed, converting a rank into a manipulable, causal mechanism (dose = vocabulary coverage, batch-shift strength, or spatial-aggregation depth).
- **Three deconfounding tools that generalize beyond genomics**—a non-linear probe with a reference-free structure metric to expose label circularity, coverage-as-measured-variable to separate representation from tokenizer, and equivalence testing to turn a null result into a quantified claim.
- **Genomics as a stress test, with two modality firsts**—four modalities, up to 24 leakage-controlled atlases, five FM families, and a 12-study meta-analysis, including the first calibration/reliability audit of a single-cell ATAC modality and the first zero-shot spatial-FM head-to-head against a parameter-free niche baseline.
- **Scope**—all findings hold in the zero-shot, frozen-embedding regime on coarse human cell types; fine-tuning and few-shot transfer are untested and can reverse them.

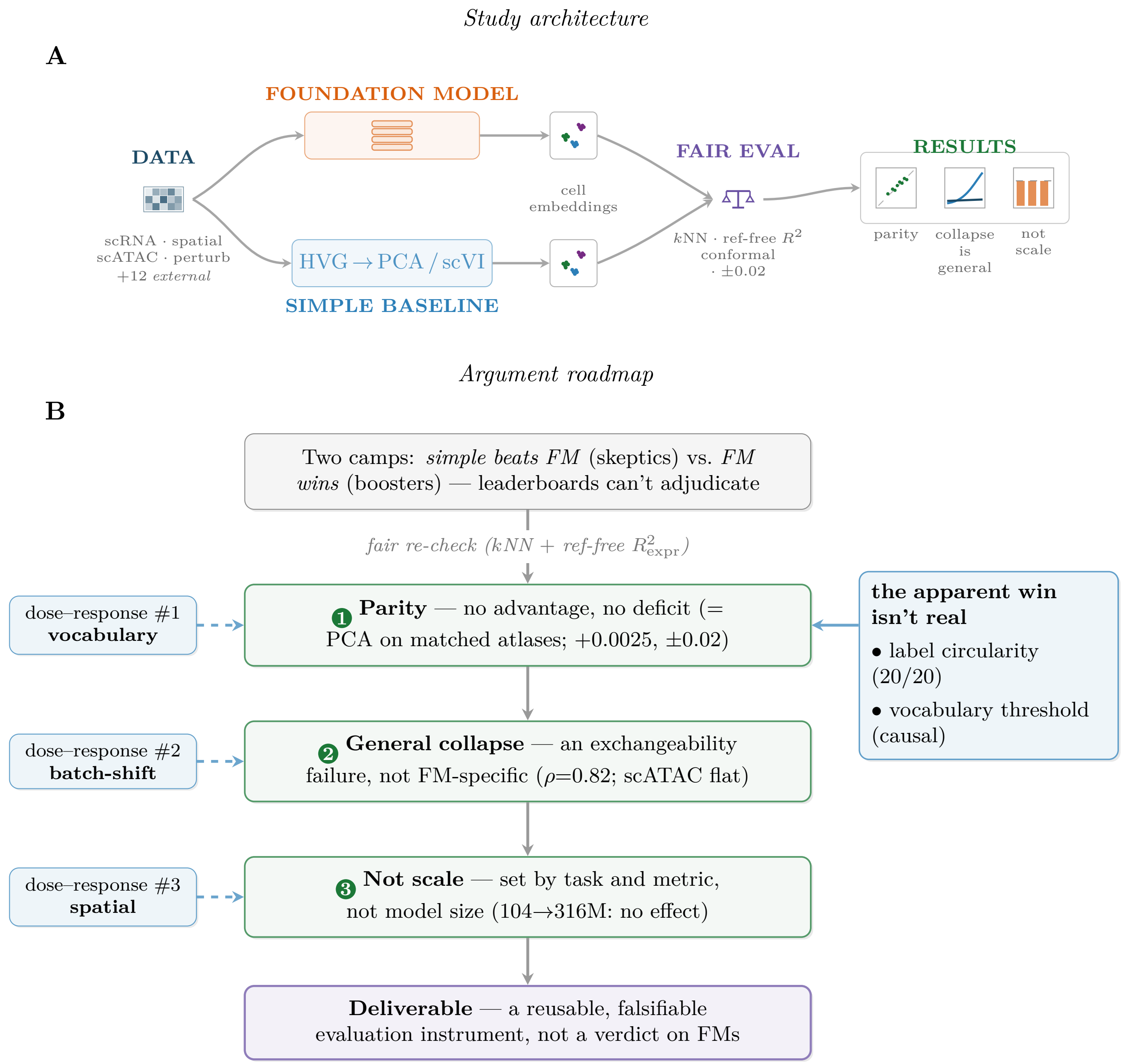


Figure 1: *Paper at a glance.* **(A)** Study architecture: four self-computed data modalities (scRNA, spatial, scATAC, perturbation) plus an external 12-study meta-analysis as context are embedded by foundation models *and* simple baselines, scored by a fair-evaluation suite (non-linear probe, reference-free structure metrics, split-conformal coverage, an equivalence test, three dose–response analyses), yielding the results at right. **(B)** Argument roadmap: a fair, reference-free re-check turns the field's two-camp dispute into three provisional findings; the apparent baseline "win" dissolves into two artifacts (label circularity and a causal vocabulary threshold), and three dose–response probes (left)—vocabulary coverage, batch-shift strength, and spatial-aggregation depth—supply the manipulable mechanisms behind the re-check.

## 2 Related work

Single-cell foundation models pretrain a transformer on millions of unlabelled cells and expose the result as a frozen feature extractor for annotation, integration, and perturbation inference [1, 2]. The first wave established the paradigm across modalities: masked- or rank-based transcriptome encoders [5–7], a cross-species protein-embedding tokenizer [8], a generative latent model [9], and spatial and chromatin variants [10–12, 18]. The implicit claim is that scale and pretraining buy a representation that recovers biology better than the largely linear pipelines it replaces. We do not train a model in this line; we take these frozen encoders as the object of measurement.

A second camp reports the opposite in the zero-shot, frozen-embedding setting, and it is with this camp that our protocol is most easily confused. Each entry is a careful but *single*-task verdict on a *single* yardstick: Kedzierska et al. [15] find zero-shot embeddings do not beat PCA on clustering-agreement metrics; Boiarsky et al. [19] show a supervised linear probe is competitive with a much larger model; Bendidi et al. [20] conclude "one PCA still rules them all" on perturbation; Atti and Subramaniam [21] argue for fundamental limitations from figure-level evidence; and Wu et al. [22] reach a biology-driven skeptical reading. The perturbation sub-literature converges independently: simple additive or linear controls match or exceed deep and foundation predictors [23–26], a pattern reproduced at benchmark scale [27–29] and against strong graph baselines [30]. Adjacent work probes why: attention captures co-expression rather than unique regulatory signal [31], and pretraining size and diversity add little [32]. We read this convergent evidence as vote-counted context [33] rather than a pooled effect, and take from it the question it leaves open—not whether a control wins but why and when. What the critiques share is a *method*: a binary rank on one agreement score, with no test of whether the score is confounded and no audit of whether the resulting confidence survives deployment.

The confound sits in the yardstick itself. Integration and annotation are scored by label-agreement suites—scIB and its ARI/NMI panels [34], consensus and hierarchical annotation pipelines [35, 36], and deep-integration benchmarks [37]—whose "ground truth" is produced by clustering and marker scoring [16, 38] in the same expression space a linear baseline occupies. Agreement with such labels rewards recovering the label-generating partition rather than the underlying signal, a circularity that overcorrection-aware batch metrics have begun to flag in a narrower setting [39]. None of the one-off critiques pairs its agreement score with a label-independent structure check, treats tokenizer coverage as a measured variable rather than a fixed assumption, or argues parity by equivalence rather than by an absent $p$-value—so a coverage artifact or a coarse-label artifact is indistinguishable from a representational deficit.

The reliability axis is almost entirely absent from both camps. Split-conformal prediction gives distribution-free coverage guarantees [40, 41], but only under exchangeability, which the batch and donor shift of single-cell data routinely breaks [17]; neural confidences are miscalibrated without correction [42]. The single-cell uptake of these tools is recent and baseline-only: conformal annotation [43] and hierarchical reject options [44] audit classical classifiers, not frozen FM embeddings. The gap widens outside transcriptomics: spatial [10] and chromatin [11, 12, 18] foundation models are evaluated almost exclusively on in-modality label recovery, and to our knowledge no calibration or conformal audit of a single-cell ATAC foundation model, nor a head-to-head of a zero-shot spatial FM against a parameter-free niche baseline, has been reported. A leaderboard rank says nothing about whether a method's uncertainty is trustworthy under shift—the property that decides whether an embedding is safe to deploy.

The missing synthesis, and our contribution, is to fold these separate observations into one reusable instrument rather than add a verdict to the pile. Where prior critiques deliver "simple beats FM" on a single task, we pair every agreement metric with a non-linear probe and a reference-free structure metric to expose circularity, promote tokenizer coverage to a measured variable, argue parity by equivalence testing, and add a reliability audit (conformal coverage, calibration, selective abstention). The move that separates us from a leaderboard is dose–response: rather than reporting that a criticized method class wins or loses, we place it *on* a curve against a manipulable cause—vocabulary coverage, batch-shift strength, spatial-aggregation depth—so a rank becomes a mechanism that can be attributed or dissolved. The framing is deliberately domain-general; single-cell genomics is the stress test that makes circularity, coverage mismatch, and exchangeability failure severe at once, not the scope of the method.

## 3 Data and methods

### 3.1 Corpus

We catalogued 37 entries (∼30 distinct benchmark studies plus infrastructure and counter-evidence); 12 contributed usable per-comparison numbers (10,759 comparisons; machine-readable result tables where released, otherwise figure/supplementary values, each source-tagged). One catalogued item was dropped as non–peer-reviewed / AI-authored (the entry and the exclusion criterion are recorded in the released extractability manifest).

### 3.2 Pooling

Within each (study, task, dataset, metric) group $g$ we pair every simple-baseline method $b$ against every FM/DL competitor $m$ and score a win indicator $w = \mathbb{1}[\, s_b \succ s_m \,]$ by the authoritative per-metric direction $\succ$ (higher- or lower-is-better as declared per metric; exact ties score 0.5). The per-study win-rate is $\hat{p}_s = \frac{1}{|C_s|} \sum_{c \in C_s} w_c$ over that study's comparison set $C_s$. Because one study (scPerturBench [28]) supplies ∼96% of raw comparisons (and >99% within the perturbation cluster), the headline estimator is the equal-weight-per-study mean $\bar{p} = \frac{1}{S} \sum_{s=1}^{S} \hat{p}_s$, with a 95% CI from a nonparametric bootstrap resampling the $S$ studies (10,000 replicates, seed fixed); the naive per-comparison pooled rate is also stored for contrast.

Method families (mean/linear/classical-DR vs. mlp/FM/DL) and metric families (correlation / MSE / distribution, split by DEG-only vs. all-genes) are tagged per row so win-rates can be stratified.

This win-rate pooling is a deliberately conservative *vote-counting* synthesis: it scores the *direction* of each comparison rather than a common effect size—which the heterogeneous metrics (ARI, MSE, AUROC, correlation) do not share—so it discards effect magnitude and per-study precision and has lower statistical power than an effect-size meta-analysis [33]. We therefore treat it as descriptive context: reported equal-weight with study-level CIs, stratified by metric family, and never the sole basis for a load-bearing claim (those rest on the self-computed re-derivations of clusters J–L, defined in Table 1).

### 3.3 Fair-evaluation core and dose–response designs

To deconfound the label-matching critique we score every representation three ways. (i) A *non-linear* probe: a distance-weighted $k$NN classifier ($k$=15) on the held-out largest batch, removing the bias of the naive comparator—a *linear* probe (multinomial logistic regression scored by one-vs-rest cross-batch macro-AUROC)—that flatters PCA (which lives in the same linear space the labels were derived from). (ii) A *reference-free* structure metric, $R^2_{\text{expr}}$ (and, for spatial niches, $R^2_{\text{compo}}$): the fraction of highly-variable-gene expression (or neighbourhood cell-type-composition) variance explained by a method's *own* unsupervised $K$-means partition (fit with $K$ equal to the number of ground-truth labels), with no labels used—so it scores structure rather than agreement with an annotation cut; where a partition is instead scored *against* labels (e.g. the spatial-niche ARI below), we use the adjusted Rand index, the chance-corrected agreement between two partitions (1 = identical, 0 = chance-level). (iii) An *equivalence* test for parity: per FM family we take the paired per-atlas difference from PCA and bootstrap its mean against a pre-stated ±0.02 margin (TOST-style), so "no advantage" is a positive equivalence result, not an underpowered failure-to-reject; per-family significance is additionally checked by a two-sided Wilcoxon signed-rank test on the paired atlas-level differences, and where multiple arms share the same atlas set the reported interval is an atlas-cluster bootstrap (resampling atlases, the unit of independence, rather than arm×atlas rows); we never summarize with a best-of-$N$ statistic, which carries a max-selection bias.

Each mechanism is then traced as a *dose–response*: apparent FM quality vs. tokenizer-coverage fraction—the fraction of an atlas's gene symbols/IDs that map onto a given FM's vocabulary, which is also the objective, performance-blind criterion used throughout to classify atlases vocabulary-matched vs. vocabulary-mismatched (near-zero mapped fraction for cross-species symbols or non-symbol gene IDs, a substantial mapped fraction otherwise)—with a causal within-atlas gene ablation that holds the atlas fixed and removes only readable genes; coverage collapse vs. a measured batch-shift strength (the cross-validated AUROC for predicting held-out-batch membership) in two degradation metrics, with the foundation models placed *on* the curve; and niche prediction vs. spatial-smoothing dose. Modality/regime contrasts (e.g. the scRNA-vs-scATAC coverage-gap slope) are tested by an interaction term in a linear regression of the gap on shift strength (a slope-difference $t$-test), corroborated by a label-permutation test on the slope difference. Spatial probes use spatially-blocked cross-validation to avoid leakage.

### 3.4 Reliability machinery

Calibration is measured by Expected Calibration Error, $\mathrm{ECE} = \sum_{j=1}^{B} \frac{|B_j|}{n} \left| \mathrm{acc}(B_j) - \mathrm{conf}(B_j) \right|$ over $B$=15 equal-width confidence bins. Readout capacity is probed with three frozen-embedding classifiers, trained on the calibration split and scored cross-batch on the held-out batch—multinomial logistic regression (linear), a 1-hidden-layer MLP-64, and MLP-256—to isolate decoder-capacity effects on out-of-batch calibration from the representation itself. Temperature scaling [42] fits a single scalar $T$ by minimizing negative log-likelihood on a held-out calibration split and rescales logits ($z/T$) before the softmax at test time. Predictive uncertainty uses split-conformal LAC sets: with calibration scores $\alpha_i = 1 - \hat{\pi}_{y_i}(x_i)$ and threshold $\hat{q} = \lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil / n_{\mathrm{cal}}$ quantile, the test set is $C(x) = \{y : 1 - \hat{\pi}_y(x) \leq \hat{q}\}$, giving marginal coverage $\geq 1 - \alpha = 0.90$ [41]; the coverage gap reported under shift is the matched in-distribution coverage minus the shifted coverage (positive = under-coverage), where the in-distribution baseline is the random/matched split—numerically the nominal 0.90 up to sampling—for cross-sample shift and the control-to-control split for cross-diagnosis (Control→AD) shift; we cross-check coverage to 16 digits against two independent conformal-prediction libraries [45, 46].

The confidence↔correctness link is a partial correlation deconfounded by log total fragments, log peak count, and sample via Frisch–Waugh–Lovell (regress confidence and correctness on the confounders, correlate residuals). Selective performance is summarized by AURC over the risk–coverage curve; negative controls shuffle labels and permute features (both must collapse to chance); all CIs are 1,000× bootstrap. Seeds fixed; all scripts deterministic.

### 3.5 Self-computed audits

Spatial niche identification on a CosMx human lymph-node spatial dataset [47] (three zero-shot spatial FMs—scGPT-spatial [48], Novae [49], Nicheformer [10]—run locally via self-written loaders that bypass uninstallable packages, with figshare/HF weights loaded exactly); scATAC cell-type calibration on GSE174367 snATAC [50] (130,418 cells, 20 samples, 7 brain types) with leakage-controlled cross-sample and cross-disease (Control→AD) splits.

The scATAC FM (ChromFound [11]) was run zero-shot from its *official* model modules and pretrained weights (loaded exactly), with only the CUDA-only Mamba selective-scan and flash-attention kernels routed to their numerically-equivalent reference implementations (the upstream Mamba reference scan and torch scaled-dot-product attention) for GPU compatibility—it is the official architecture, not a re-implementation—on a disclosed top-2048-peak / 20k-cell subset for tractability. A second scATAC FM, Atacformer [12] (a standard transformer encoder over an 890kregion hg38 universe), is added via a self-written loader that maps GSE174367 peaks to universe tokens by genomic overlap (191,858/219,070 peaks mapped; weights loaded 0 missing/0 unexpected); EpiFoundation [18] remains unrun (env-incompatible kernels), disclosed as a limitation.

The self-computed scRNA suite (clusters J–L) uses local leakage-controlled atlases with frozen FM embeddings, classical baselines (PCA/HVG/scVI [13]) computed locally, leave-one-batch-out splits, and the same calibration/conformal/abstention machinery; batch integration is scored by LISI (the local inverse Simpson's index over each cell's $k$-nearest-neighbour graph at a fixed neighbourhood size/perplexity), with two label variants: iLISI uses batch identity (higher = better batch mixing) and cLISI uses cell-type identity (higher = more cell-type mixing, i.e. worse purity). The perturbation arm uses GEARS [30] on Norman-2019 [51], benchmarked against an additive baseline that predicts control $+ \delta_A + \delta_B$ from train-only single-perturbation deltas $\delta_A, \delta_B$; double-perturbation combinations are stratified by combo novelty—seen2/seen1/seen0, i.e. both/one/neither constituent single perturbation observed in training—and scored by full-gene MSE (mean squared error over all measured genes, with no oracle/DE gene subselection).

The expansion sweep runs an enlarged simple-baseline panel (adding $k$NN, nearest-centroid, random-forest) across 24 annotated atlases drawn from a 95-atlas local inventory (selected for cell-type + a batch field with $\geq$2 levels), and adds further self-embedded foundation models zero-shot—Geneformer-V2-104M (the package is bypassed: faithful rank-value tokenization—normalize-to-10k / gene-median, rank-descending, a class token, the expressed genes, and an end token—and mean-pooling over gene tokens), and, for the fair multi-FM check, four more arms adding three further families: Geneformer-V2-316M, scFoundation (Performer masked-autoencoder, vendored model code), CellPLM (flowformer+GMVAE, vendored package), and UCE (protein-embedding tokenizer, driven via its released eval pipeline).

The FM arm is grown from 3 to 20 raw-count atlases by transferring labels from each curated (normalized) atlas onto its raw-count GSE twin via cell-barcode matching; baseline representations are recomputed on log-CP10k-normalized counts for fairness. Seeds fixed; thread-capped for determinism.

### 3.6 Reproducibility

All results derive from deterministic scripts (fixed seeds) over released result tables and local data; conformal coverage agrees to 16 digits across a custom implementation, an independent re-derivation, and two public conformal-prediction libraries [45, 46]; self-computed audits passed independent adversarial verification, which corrected an initial scATAC-FM overclaim into the matched-control statement reported here. The two most-relied-on scRNA FM loaders are additionally validated against official model code: scGPT reproduces the official cached embedding to three decimals, and Geneformer-V2's loader reproduces the official geneformer tokenizer exactly (cosine 1.000 on bone marrow; the official 4096-token budget gives the same $k$NN-AUROC as our 2048, 0.872 vs. 0.871), confirming the tokenization is faithful and the shorter budget pessimistic-for-the-FM. The scATAC Atacformer loader was checked the same way against the official geniml model code: the released architecture is a plain transformer encoder (no positional embeddings), which our loader reconstructs; this surfaced one bug—a GeLU feed-forward where the official uses ReLU—that shifts the embedding (cosine 0.93) but not the cross-sample cell-type accuracy (0.962 with the official ReLU vs. 0.965 ours, both below LSI). Because the official class does not instantiate on the current transformers release (an upstream API change), we ran a faithful replica of its architecture loaded with the official weights.

We assign every self-computed scRNA claim (clusters J–L) a confidence level—supported, qualified, or withheld—and phrase it with correspondingly conservative wording; for example, we write "residual benchmark-correctness signal" rather than "biological signal" and "cross-batch/exchangeability gap" rather than "leakage gap." Under this convention, cluster J is reported as reproduction (qualified for the second FM), cluster K's coverage-collapse is reported as its best-supported result while its confidence-signal wording is scoped to benchmark correctness, and cluster L's additive-beats-GEARS result is now supported by a label-fair full-gene comparison (superseding the earlier oracle-gene version), with only the selective-prediction/uncertainty half still exploratory (magnitude-deconfounded). One earlier intermediate result, affected by a parser bug, was flagged during this verification and is excluded here. A provenance manifest accompanies the released code: the SHA-256 of every paper-backing result table and every first-party script, plus the captured runtime environment (Python and package versions), so any result table can be checked bit-for-bit against a re-run of its producing script.

## 4 Results

Results are cross-indexed by *cluster* (A–L), each mapping an external or self-computed arm to its task; the full mapping is in Table 1, and the self-computed clusters are H–L (the scRNA suite is J–L).

Table 1: Cross-cluster summary. Win-rate is the equal-weight per-study mean (95% CI by study-level bootstrap); $k$ = number of studies. Clusters are kept separate (different tasks/metrics; not pooled into a single number). E and F are one merged GRN cluster, so the twelve labels A–L are eleven clusters; rows follow the external-vs-self-computed grouping of the footnote, not alphabetical order.

| Cluster | Claim (zero-shot) | Win-rate [95% CI] | Strength |
|---|---|---|---|
| A | no FM advantage (rep./annot./integ.) | 0.887 [0.71, 1.00], $k$=5 | circular-metric† |
| B | linear/additive $\geq$ DL/FM (perturbation) | 0.653 [0.37, 0.87], $k$=7 | metric-contingent |
| H | no spatial-FM advantage (niche-ID), fair test | FM kNN 0.76–0.86 | circular-metric† |
| E/F | raw scFM signal $\leq$ gene-level baselines (GRN) | 3/3, $k$=1 | usage-dependent |
| G | trajectory/velocity: no scFM evaluated | — | open gap |
| D | scIB/kBET/LISI unreliable at scale | — | caveat |
| C | ensemble/calibration give reliable UQ; FM UQ untested | — | supportive |
| I | scATAC calibration robust to shift; 2 FMs, no gain over LSI | self-computed, 2 FMs | first audit |
| J | zero-shot scFM = baseline (parity), fair test | FM ties/beats PCA, 11 matched | reproduction+ |
| K | cross-batch coverage collapse is general (not FM-specific) | 20/24 atlases | novel (reframed) |
| L | additive > GEARS (perturbation, fair full-gene); FM UQ modest | 3/3 strata | supported |

Clusters A–G aggregate or audit external results; H and I are self-computed audits (spatial, scATAC) and J–L are self-computed scRNA reproductions/audits on local atlases. †Clusters A and H rest on label-matching metrics (ARI/scIB, and spatial-niche ARI) with a circularity our fair re-checks expose (via cluster J and cluster H's own composition re-check, respectively); read as "no FM advantage," not a decisive baseline win.

### 4.1 Representation, annotation, integration: no FM advantage

Across five studies [15, 19–22], simple baselines beat zero-shot scFMs in 88.7% of studies (equal-weight per study; 84.6% of the 331 individual comparisons; 95% CI [0.714, 1.00]); for classical dimensionality reduction (HVG/PCA/scVI/Harmony) the win-rate is 96.7% [0.901, 1.00] (Fig. 3). One of these five—Atti & Subramaniam [21] (entry A4)—is a tier-C source scored *qualitatively* (from figures/abstract) without extractable per-comparison values, so it enters the equal-weight mean as a single 1.00 study; dropping it barely moves the cluster-A win-rate (0.887→0.86), so the conclusion does not rest on it, and all pooled values are re-used from each study's public release (no re-analysis of raw data).

*An important caveat applies to these headline numbers and to the underlying literature:* they are almost all label-matching metrics (ARI, NMI, scIB) scored against clustering- or marker-derived cell-type labels that live in the same expression space as the linear baselines. Our self-computed re-analysis (cluster J, Fig. 2) shows *two* artifacts inflate the apparent baseline margin: label-matching circularity, and—separately—gene-vocabulary mismatch, where cross-species (mouse-symbol) and gene-ID-format (ENSG) atlases are scored as FM failures even though the human-symbol tokenizer maps almost none of their genes. Once both are controlled, on vocabulary-matched atlases the baseline and the FM are *tied* (parity on a fair non-linear probe and a reference-free structure metric; Table 4).

The defensible reading of cluster A (representation/annotation/integration) is therefore no FM advantage and no FM deficit (zero-shot scFMs neither beat nor fall behind far cheaper baselines on vocabulary-matched coarse cell types), rather than a decisive baseline victory.

### 4.2 Perturbation prediction: real but metric-contingent

Pooled equal-weight win-rate is 65.3% [0.372, 0.872], falling to 51.8% [0.151, 0.867] when restricted to studies with $\geq$3 comparisons. Both intervals are wide and *include parity (0.50)*—cluster B (perturbation) rests on only $k$=7 studies (4 in the restricted stratum), so this is an under-powered "no clear winner" rather than an established baseline advantage. Baselines win on delta/$L_2$/distribution metrics (Csendes [25], Kernfeld [24], Wong–Pfizer [23] at 1.00; Ahlmann-Eltze [26] 0.60) but FMs/DL are competitive on absolute-correlation metrics (PertEval [27] 0.00). The contingency is the finding: which side "wins" depends on the metric family.

### 4.3 Spatial niche identification: no FM advantage after a fair re-check

On the CosMx lymph node (19,718 cells, 4 niches) [47], an unsupervised clustering scored by ARI against the niche labels makes three zero-shot spatial FMs look near-useless: ARI 0.386 (nbhd-composition baseline) vs. scGPT-spatial 0.139, Novae 0.124, Nicheformer 0.012. The same fair re-check we applied to cluster J overturns most of that gap (Fig. 4). First, the task is *composition-defined*: the named niches (B-/T-cell zone, germinal center, medulla) explain abundant neighborhood cell-type-composition variance ($R^2_{\text{compo}}$ 0.234) but essentially no single-cell expression variance ($R^2_{\text{expr}}$ 0.007), so a per-cell expression FM is scored against a label with no per-cell expression signal, and the nbhd-composition baseline—which *is* the niche-definition recipe ($R^2_{\text{compo}}$ 0.254≈0.234)—wins near-tautologically.

Second, a *supervised* $k$NN niche-probe (spatially-blocked CV) shows all three FMs in fact carry substantial niche information—AUROC 0.86 (Novae), 0.83 (scGPT-spatial), 0.76 (Nicheformer)—far above what their 0.01–0.14 ARI implied; the unsupervised ARI was largely a clustering/metric artifact.

A *residual, real* gap remains: even supervised, the FMs (0.76–0.86) trail the simple spatial baselines (spatial-smoothed PCA 0.92 at a fixed smoothing radius, nbhd-composition 0.90; the dose–response of Fig. 5 traces the full smoothing curve, which peaks slightly higher at AUROC 0.93). The verdict is therefore no FM advantage—simple neighborhood/spatial baselines match or beat the spatial FMs on a composition-defined task—rather than the original "FMs are near-useless," which the ARI metric over-stated.

*A spatial dose–response localizes the residual gap to one axis: spatial context.* If the niche task is composition-defined, then the only thing a per-cell expression model lacks is spatial aggregation—so we dose exactly that. Taking a purely per-cell representation (PCA on expression) and progressively spatial-smoothing it over its $k$ nearest spatial neighbours, niche prediction rises monotonically with the dose (Fig. 5). At $k$=0 (per-cell) it sits at the FM level (AUROC 0.835, macro-F1 0.40—compare

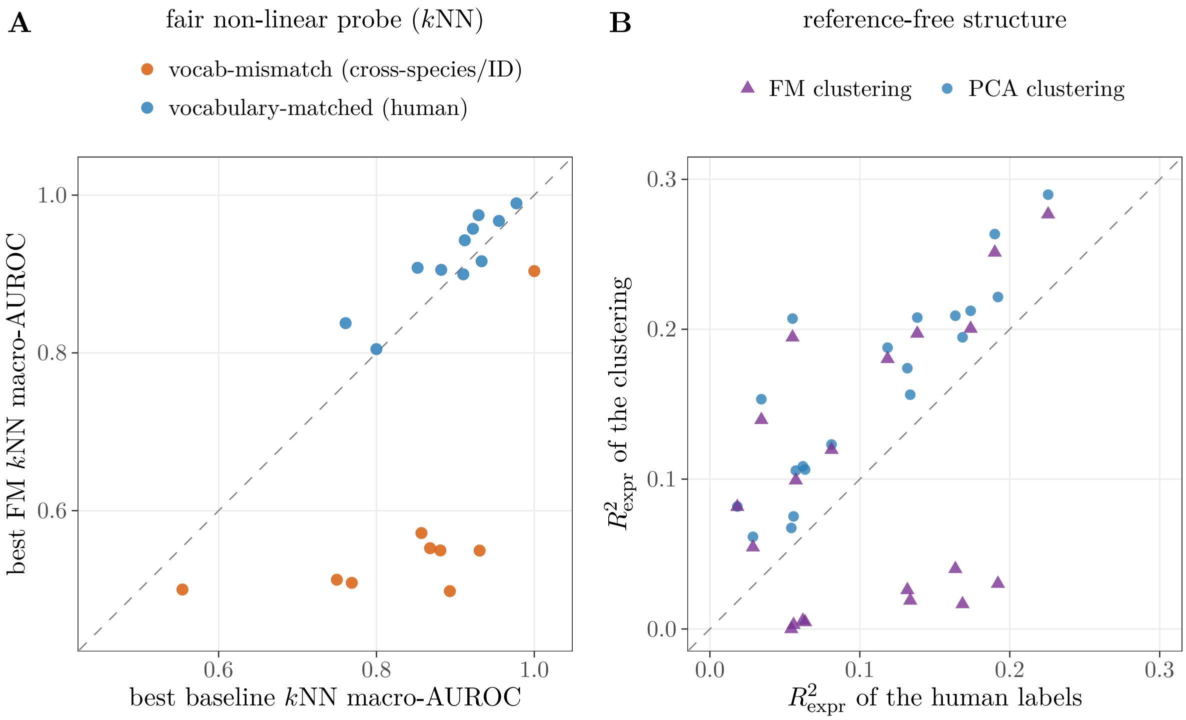


Figure 2: *The corrected view* (cluster J). **(A)** a fair non-linear ($k$NN) probe over six FM arms across five families (scGPT, Geneformer-V2 104M/316M, scFoundation, CellPLM, UCE; best per atlas). On the 11 blue *vocabulary-matched* human atlases the best-per-atlas FM sits on/near the diagonal (this best-of view carries a +0.024 max-selection edge over PCA; unbiased parity—tie band $|\Delta| < 0.02$—is the per-*family* equivalence test of Table 4, not these best-of points); the off-diagonal floor is the 8 cross-species (mouse-symbol) orange atlases, where the human FM tokenizer maps almost no genes ($k$NN≈0.53)—a tokenization handicap, not a representational failure; the 9th orange (vocab-mismatch) atlas is the ENSG-only one, and it sits near PCA parity at top-right precisely because its best FM (Geneformer-V2) still maps ∼24% of its genes—the same coverage effect, read the other way. **(B)** reference-free $R^2_{\text{expr}}$—every point above the diagonal means the human cell-type labels explain *less* HVG-expression variance than the PCA clustering (20/20 atlases; than both clusterings in 11/20): the labels are a *coarser* partition than unsupervised clustering recovers, so agreement-with-label scores reward reproducing that coarse cut rather than the finer expression structure—corroborating the circularity rather than proving it alone (the direct evidence is the linear-vs-non-linear probe gap). Conclusion: once vocabulary mismatch and label circularity are separated out, zero-shot scFMs *neither beat nor trail* PCA on coarse human cell types.

raw-FM AUROC scGPT-spatial 0.826, Novae 0.862), and as $k$ grows it climbs to AUROC 0.93 / F1 0.76 (Spearman $\rho = 0.98$ for F1), *overtaking all three spatial FMs and the neighbourhood-composition baseline* by $k \approx 25$–50 before over-smoothing erodes it past $k$=100.

The same dose lifts an FM representation too (spatial-smoothing scGPT-spatial improves its F1 by +0.20, $\rho = 0.93$), confirming the missing ingredient is shared. So the residual “spatial-FM gap” is not a representational deficit but a single missing operation—spatial aggregation—that a trivial smoothing of per-cell PCA supplies. The spatial FMs, scored per-cell, are simply not given the dose the task rewards.

A *differential* check supports the reading: spatial smoothing helps the purely per-cell PCA most (macro-F1 gain +0.36) and helps the strongest raw spatial FM least—Novae, the only FM with high

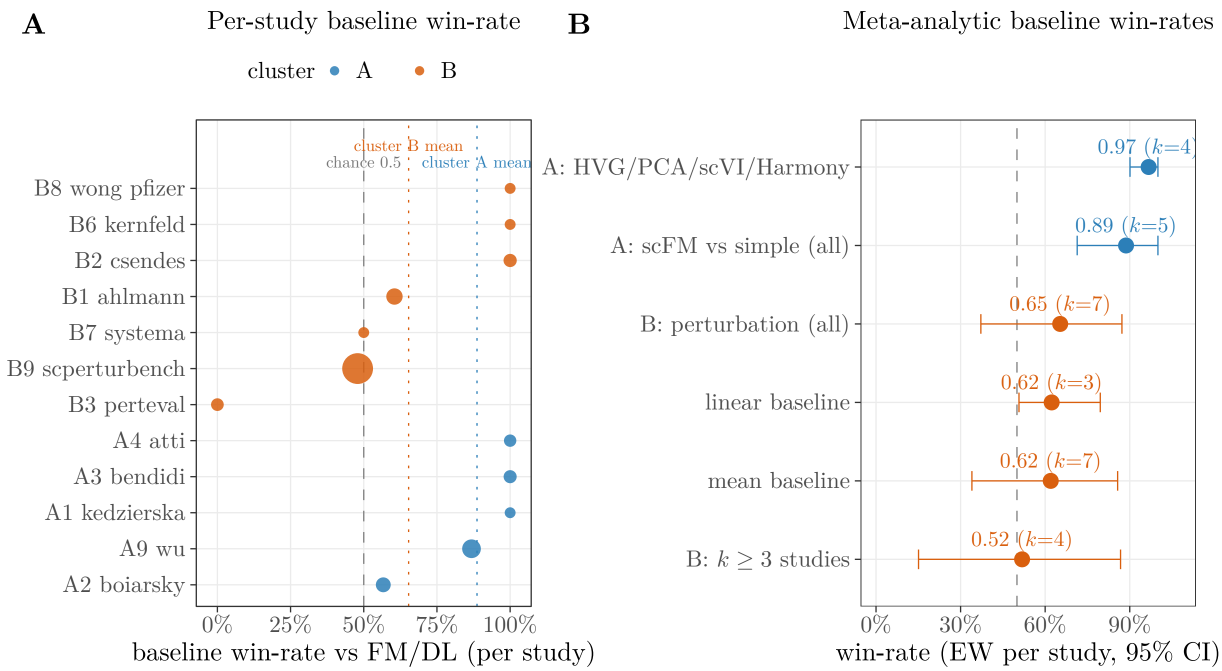


Figure 3: *External meta-analysis (clusters A–B).* **(A)** Per-study baseline win-rate vs. FM/DL (point size $\propto$ #comparisons). **(B)** Meta-analytic baseline win-rates by stratum (equal-weight per study; 95% CI over studies).

per-cell niche-F1 (0.53), gains just +0.06—consistent with Novae already encoding neighbourhood context and PCA lacking it entirely. The residual is the aggregation a per-cell view omits.

(The pattern is not perfectly monotone across FMs—Nicheformer, a weak raw embedder, still gains +0.28—so we report it as supportive, not decisive. A second, unsupervised mechanism check we ran was *null*: the smoothed-PCA *clustering*'s composition-alignment $R^2_{\text{compo}}$ does not rise with the dose (it peaks at $k$=6 then declines, $\rho = -0.41$, as large-$k$ over-smoothing blurs the partition), so we rest the claim on the supervised probe, not on partition structure.)

This is a mechanism demonstration on a single dataset (one CosMx lymph node, eight dose levels), and the smoothing graph is built over all cells, so a small fraction of each test cell's smoothing neighbours fall in adjacent CV folds ($\leq$ 6% at the largest $k$, $\leq$ 3% where the curve overtakes the FMs); a strictly fold-blocked recompute (smoothing restricted to same-fold neighbours, no cross-fold edges) preserves the overtake—smoothed-PCA reaches macro-F1 0.70 versus the raw spatial FMs' 0.34–0.53 and the composition baseline's 0.54—so the result is not a smoothing-leak artifact. The single dataset still bounds its generality.

### 4.4 GRN, trajectory, integration metrics, and uncertainty

*GRN:* raw scFM signal (Geneformer attention AUROC 0.70) loses to gene-level baselines (variance 0.88, mean-expression 0.84, 1−dropout 0.81; 3/3) [31]; FM embeddings help only when fed to a *supervised* GNN (scRegNet [52]; usage-dependent). *Trajectory/velocity:* a 15-method×17-dataset benchmark [53] contains *zero* foundation models—an open modality gap. *Integration metrics:* kBET/iLISI discriminatory power collapses to zero under large batch effects and scIB [34] is gameable [37, 39], which caveats all scFM scIB comparisons (including ours). *Uncertainty:* ensemble-

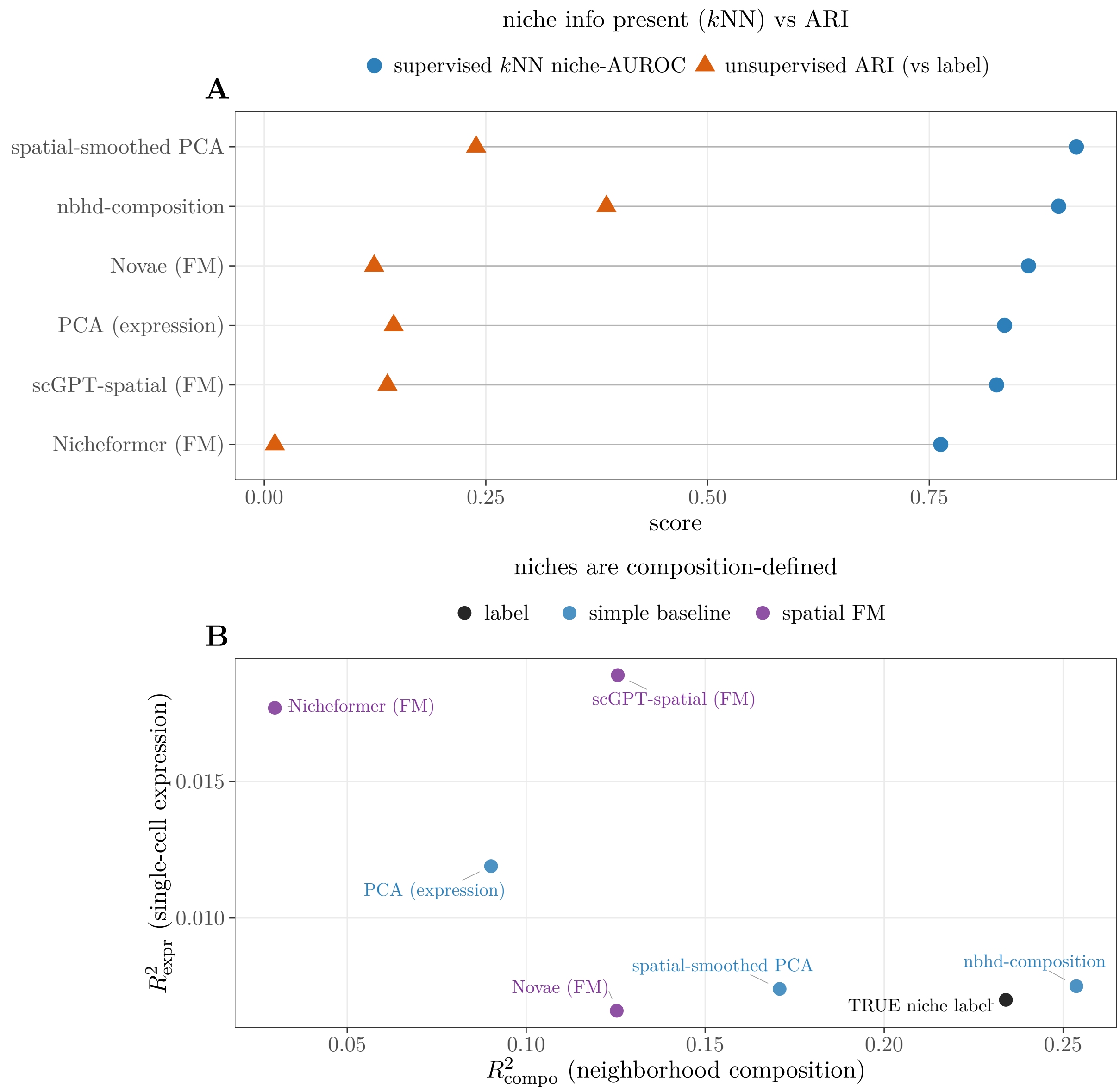


Figure 4: Cluster-H fair re-check (spatial niche-ID). **(A)** unsupervised ARI-vs-label (orange) says the spatial FMs fail, but a supervised $k$NN niche-probe (blue) shows all three carry niche information (AUROC 0.76–0.86); the residual gap to spatial baselines is small. **(B)** the niche labels are composition-defined ($R^2_{\mathrm{compo}}$ 0.234) with near-zero single-cell expression signal ($R^2_{\mathrm{expr}}$ 0.007), so the nbhd-composition baseline winning ARI is near-tautological. Same circularity as cluster J.

consensus and calibration give reliable uncertainty (e.g. popV [35] consensus score 8→95% correct; hierarchical-loss annotation with a reject option [36, 44]), but FM-specific uncertainty remains unaudited.

### 4.5 scATAC-seq reliability: a first audit and a modality contrast

The FM-free peak-LSI cell-type probe is excellently calibrated (ECE 0.006) and its split-conformal coverage holds under both cross-sample (gap −0.007) and cross-disease Control→AD shift (gap

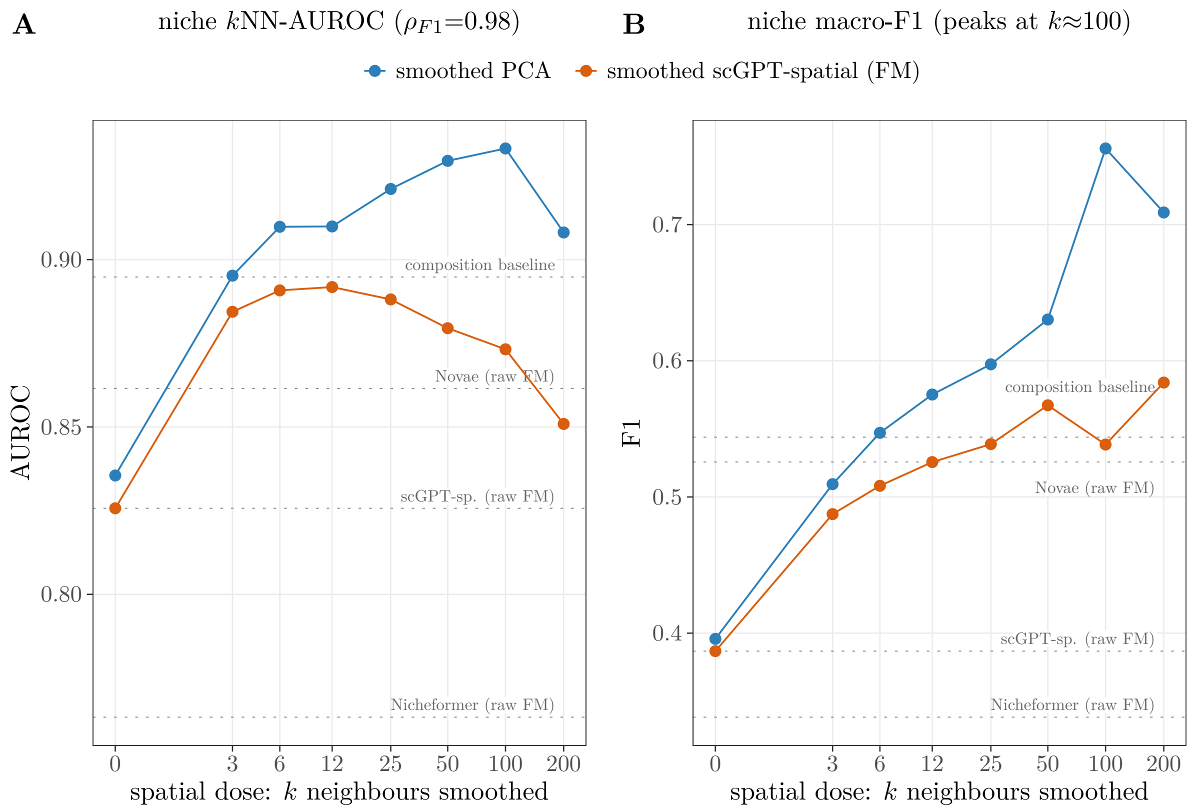


Figure 5: *Spatial dose–response* (depth probe for the residual niche gap). Per-cell representations are progressively spatial-smoothed over $k$ nearest spatial neighbours ($x$, log scale; $k$=0 is per-cell); $y$ is supervised niche prediction under spatially-blocked CV—**(A)** $k$NN macro-AUROC and **(B)** macro-F1. Smoothing per-cell PCA (blue) lifts niche prediction monotonically (F1 Spearman $\rho = 0.98$) from the per-cell-FM level past all three raw spatial FMs and the composition baseline (dotted references) by $k \approx$ 25–50; the same dose helps a smoothed FM (orange) too. The residual "spatial-FM gap" is therefore the spatial-aggregation axis—one operation a per-cell model lacks—not a representational deficit.

−0.028)—in contrast to the scRNA scGPT regime, where cross-batch coverage collapses by 0.123 (Fig. 6). Confidence carries a depth-independent correctness signal (partial $r = 0.478 \approx$ raw). This is, to our knowledge, the first calibration audit in a single-cell ATAC modality; the closest prior work applies conformal annotation to scRNA only [43], without a frozen-FM target, a coverage-gap-under-shift measurement, or depth deconfounding. (The circular-label concern of cluster A applies to the *accuracy* here too, but the load-bearing finding—whether split-conformal *coverage* holds under shift—is a property of cal/test exchangeability that is invariant to the label definition and is computed identically for every method, so the modality contrast below is robust to it.)

We then ran the first audit of *real* scATAC FMs—and, to avoid resting on a single model, we audited two independent ones (Table 2, Fig. 6B,C). ChromFound's zero-shot embedding does *not* beat its own raw TF-IDF input (accuracy 0.834 vs. 0.850; ECE 0.094 vs. 0.088). Atacformer [12] (databio; a standard transformer encoder over an 890k-region genome universe, run via a self-written loader that maps GSE174367 peaks to universe tokens by genomic overlap, and run without positional embeddings, matching its released code—a plain transformer encoder, which our loader reconstructs; running the official architecture against ours showed our loader used a GeLU

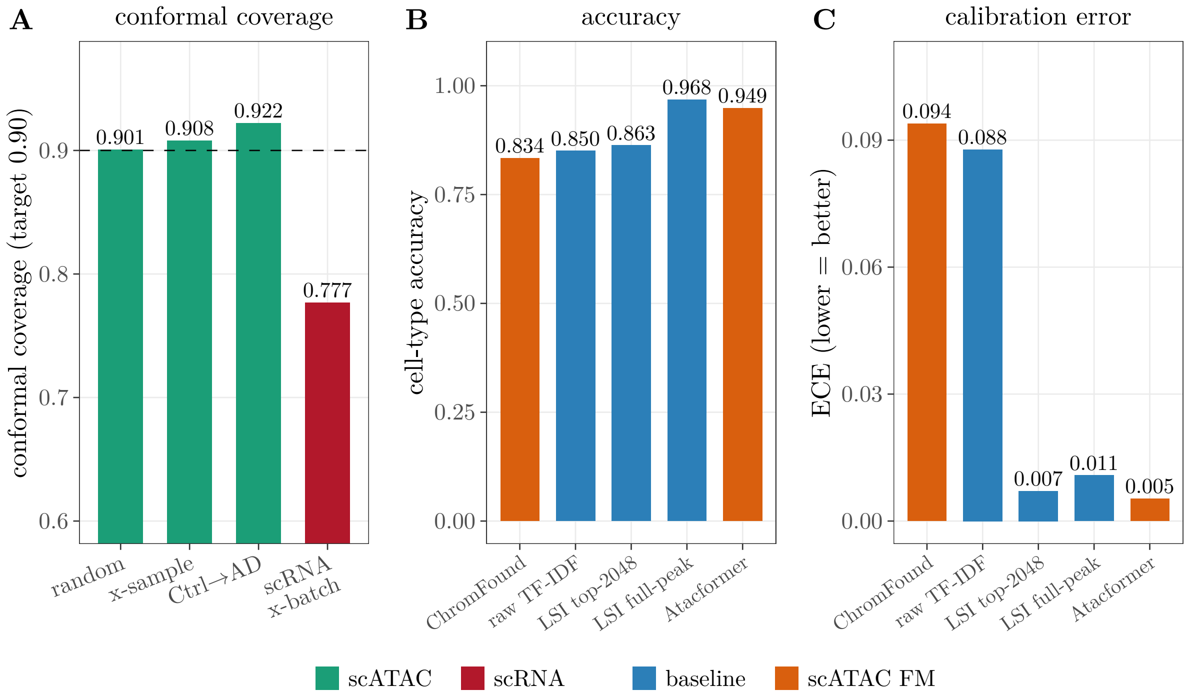


Figure 6: *scATAC (cluster I).* **(A)** Conformal coverage holds under cross-sample and cross-disease shift, whereas the scRNA scGPT regime collapses. **(B,C)** Matched comparison of two scATAC FMs against simple LSI baselines: neither FM beats full-peak LSI on accuracy, and the ECE difference is a representation-dimensionality effect.

feed-forward where the official uses ReLU, but re-embedding with the official ReLU changes the embedding only modestly (cosine 0.93) and leaves the cross-sample cell-type accuracy unchanged (0.962 vs. our 0.965, both on the loader-verification split; the matched-peak head-to-head below uses a different split and reports 0.949), so the comparison below is robust to the reconstruction) is a stronger embedding—accuracy 0.949, ECE 0.005—yet still does *not* beat the simple full-peak LSI baseline (0.968).

The two FMs together confirm that calibration tracks dimensionality, not FM-ness: the 192-dimensional Atacformer is as well-calibrated as a 50-dimensional SVD (ECE 0.005 vs. 0.007), whereas any 2048-dimensional representation (ChromFound or its raw input) is not ($\sim$0.09). For the modality contrast, conformal coverage is robust for *both* FMs and *both* baselines—no collapse under cross-sample (Atacformer gap $-0.010$) or cross-diagnosis Control→AD shift (gap $-0.013$), with a meaningful depth-independent confidence signal (partial $r = 0.491$) and negative controls at chance. The conclusion: across two scATAC FMs, the FM provides no accuracy gain over simple baselines, but scATAC calibration/coverage is genuinely robust to shift—unlike the scRNA regime, a contrast we make quantitative on a common batch-shift ruler in cluster K (cross-batch coverage collapse; Fig. 7B): the scATAC coverage gap is flat in batch-shift ($\rho = -0.24$, *n.s.*) where the scRNA gap climbs steeply ($\rho = 0.82$); the two slopes differ sharply (modality×shift interaction $p = 1.6 \times 10^{-5}$, a within-dataset contrast on the single scATAC atlas—quantified in cluster K).

Table 3 gives the full reliability profile (calibration, three coverage regimes, depth-deconfounded confidence, selective AURC, rare-type bias, negative controls) for all three representations: this robustness is label-independent (negative controls collapse to chance or below—ChromFound's 2048-d shuffle drops under it), making it the constructive counterpart to the scRNA collapse of cluster K.

Table 2: scATAC matched comparison (same 20,002 cells). Two independent scATAC FMs: neither beats the simple full-peak LSI baseline; calibration (ECE) tracks *dimensionality*, not FM-vs-baseline (the 192-d Atacformer is as well-calibrated as the 50-d SVD, while the 2048-d ChromFound is not); conformal coverage is robust for all.

| Representation | dim | accuracy | ECE | cross-sample coverage |
|---|---|---|---|---|
| ChromFound FM (faithful recipe) | 2048 | 0.834 | 0.094 | 0.904 |
| raw top-2048 log-TF-IDF (ChromFound's own input) | 2048 | 0.850 | 0.088 | 0.909 |
| **Atacformer FM** (full-universe, self-written) | 192 | 0.949 | **0.005** | 0.910 |
| LSI top-2048 → SVD-50 (matched peaks) | 50 | 0.863 | 0.007 | 0.903 |
| LSI full-peak → SVD-50 (reference) | 50 | **0.968** | 0.011 | 0.914 |

Table 3: scATAC reliability profile (GSE174367; the FM-free peak-LSI column is the full 130,418-cell audit, the two scATAC FMs a matched 20,002-cell subset): the FM-free peak-LSI probe and *both* scATAC FMs are well-calibrated and keep conformal coverage under cross-sample and cross-disease (Control→AD) shift, with a depth-deconfounded confidence signal and negative controls at or below chance—the positive, label-independent counterpart to the scRNA cross-batch collapse (cluster K). Rare-type (pericyte/endothelial) over-rejection is the shared fairness cost. (Accuracy/coverage here are from the full reliability audit; the matched-peak head-to-head of Table 2 uses a separate split, so its point estimates for the same baselines differ slightly.) The coverage gaps quoted in the text are the matched in-distribution coverage minus the shifted coverage (the random split for cross-sample, the Control→Control split for Control→AD).

| | FM-free peak-LSI | ChromFound FM | Atacformer FM |
|---|---|---|---|
| cell-type accuracy | **0.983** | 0.834 | 0.949 |
| ECE | 0.006 | 0.094 | 0.005 |
| coverage, random | 0.901 | 0.899 | 0.900 |
| coverage, cross-sample | 0.908 | 0.912 | 0.910 |
| coverage, Control→Control (in-dist) | 0.894 | 0.910 | 0.919 |
| coverage, Control→AD | 0.922 | 0.927 | 0.932 |
| confidence partial-$r$ (deconfounded) | 0.478 | 0.438 | 0.491 |
| AURC (selective) | 0.001 | 0.039 | 0.006 |
| rare-type (pericyte/endothelial) rejection | 0.58 | 0.62 | 0.38 |
| neg-control (label shuffle; chance 0.45) | 0.45 | 0.24 | 0.45 |

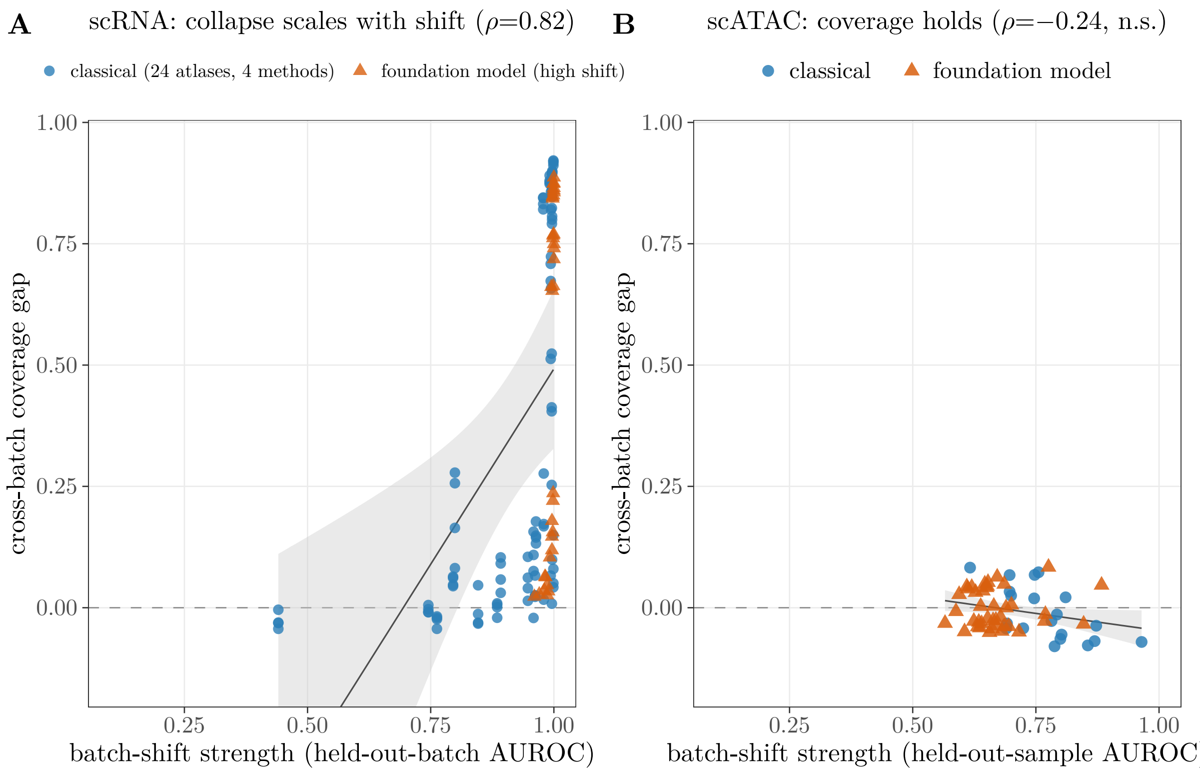


Figure 7: *Batch-shift dose–response and modality contrast, on one ruler.* $x$ is batch-shift strength (cross-validated AUROC for predicting held-out-batch/-sample membership; 0.5 = exchangeable, 1.0 = fully separable); $y$ is the cross-batch conformal coverage gap; axes shared. **(A)** (scRNA): the blue points are the 24-atlas × 4-classical-method sweep (full shift range 0.13–1.0); the fitted curve and $\rho = 0.82$ are PCA+logreg; the three FM families (scGPT, Geneformer-V2-104M, scFoundation, orange), run on the vocabulary-matched atlases, reach only the high-shift end (shift $\geq 0.96$) and there collapse by the same amount as classical (+0.48 vs. +0.49), so FMs do not repair the collapse. **(B)** (scATAC, same axes): the identical protocol on peak-LSI plus two scATAC FMs (ChromFound, Atacformer) gives a *flat* curve—coverage gap near zero across the whole shift range ($\rho = -0.24$, *n.s.*; mean −0.005, max +0.08). The scRNA collapse also holds for a second degradation metric (cross-batch ECE gap, $\rho = 0.76$) and a classifier-free shift measure (standardized centroid distance, $\rho = 0.58$–$0.60$); the 24-atlas × 4-classical-method breadth ($\rho = 0.75$) is in the text. Coverage collapse is thus a graded property of the scRNA cross-batch regime—shared by every classical method and foundation model there, and absent in scATAC.

### 4.6 Self-computed scRNA suite: representation (cluster J)

Beyond aggregating external results, we independently re-derived the core claims on local, leakage-controlled scRNA atlases (frozen zero-shot embeddings; supervised leave-one-batch-out readouts; deterministic seeds); every claim below is reported at the confidence level supported by adversarial verification (Sec. 3.6). On four atlases (lung-GEO, B-cells, cardiac-lymph, intestine; 4–26 donors) under leave-one-batch-out, frozen scGPT does *not* dominate classical baselines: PCA or scVI [13] is the top method in every atlas and scGPT is never best (Fig. 9). On the lung atlas, scGPT is worst in 4/4 folds with non-overlapping ranges (scGPT $\leq 0.961 <$ PCA $\geq 0.979$). A second, architecturally distinct FM (Geneformer) is the weakest of all in 3/3 atlases. On batch *integration* the FM advantage is axis-specific: scGPT mixes batches better (higher iLISI in 3/3) but blurs cell types (worse cLISI; cardiac 5.33 vs PCA 2.20), and the purpose-built scVI dominates the mixing/purity trade-off (Fig. 8). This is a zero-shot/frozen-readout benchmark observation reproducing the external cluster-A literature, *not* a universal FM-failure claim.

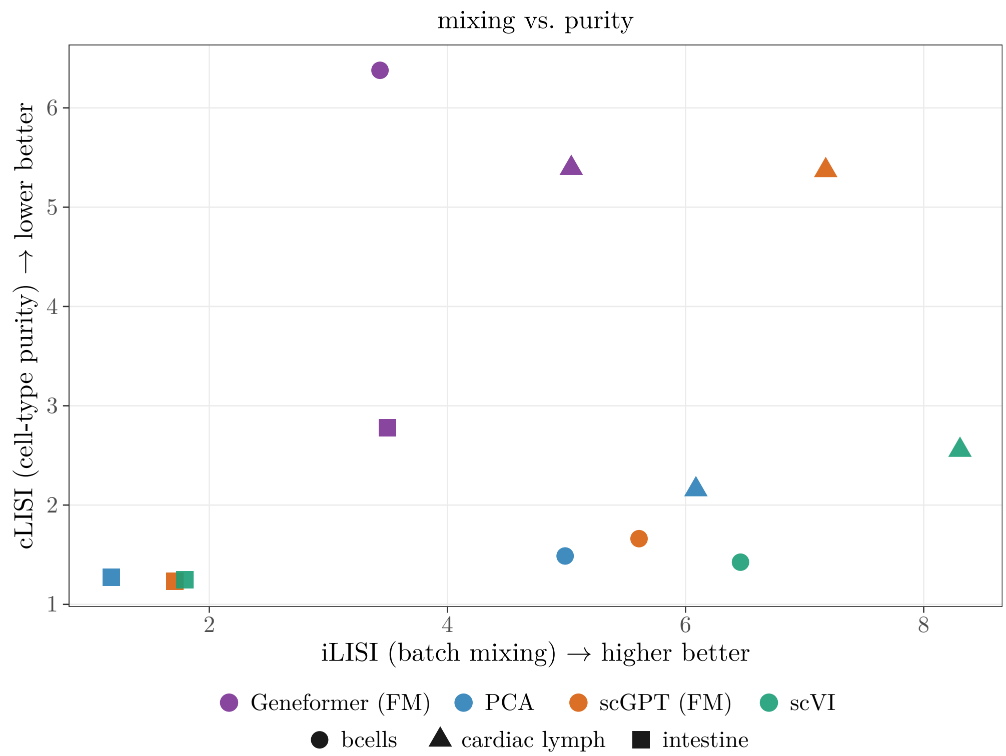


Figure 8: Integration trade-off (cluster J). The one axis on which foundation models "win"—batch mixing (higher iLISI)—comes at a cell-type-purity cost (higher cLISI = more blurring); the purpose-built scVI dominates the mixing/purity trade-off (strictly best on both axes for B-cells, and on the Pareto front for the other two atlases), so even the FM's apparent integration advantage is not an all-things-considered win. Applying the same logic we use on the FM *losses*, the FM mixing gain is regime-coupled to a cost on all three atlases (the per-atlas best-mixing FM shows an iLISI advantage +0.6/+1.1/+2.4 each paired with a cLISI cost +0.1/+3.1/+1.5); with only three integration atlases this is an observation, not a dose–response, but it shows the lone FM "win" is no freer of confounds than the apparent "losses."

We then *expanded* this audit on all three axes. *Data:* we added further foundation models (six arms across five families) and grew the FM-vs-baseline comparison from 3 to 20 raw-count atlases by transferring cell-type labels from the curated (normalized) atlases onto their raw-count twins via cell-barcode matching ($\geq$50% barcode overlap; typically 100%), recovering raw counts needed for FM tokenization. *Baselines:* an enlarged panel (PCA+logreg, HVG+logreg, $k$NN, nearest-centroid, random-forest), with the reliability sweep also run across 24 atlases (cluster K). *Models:* six FM arms across five families run zero-shot on the 20 atlases via self-written loaders (the four added later—Geneformer-V2-316M, scFoundation, CellPLM, UCE—enter the fair multi-FM check below); the two primary embedders are the newly downloaded Geneformer-V2-104M (faithful rank-value tokenization, mean-pooling over gene tokens) and scGPT-human (normalize/log1p/51-bin value-binning, class-token pooling; our self-written embedder reproduces the official cached scGPT embedding to three decimals, AUROC 0.974 vs. 0.974).

A naive scoring—cross-batch macro-AUROC of a *linear* probe against the cell-type labels—makes a baseline "match or beat" at least one FM in 16/20 atlases (and *both* FMs in 14/20;

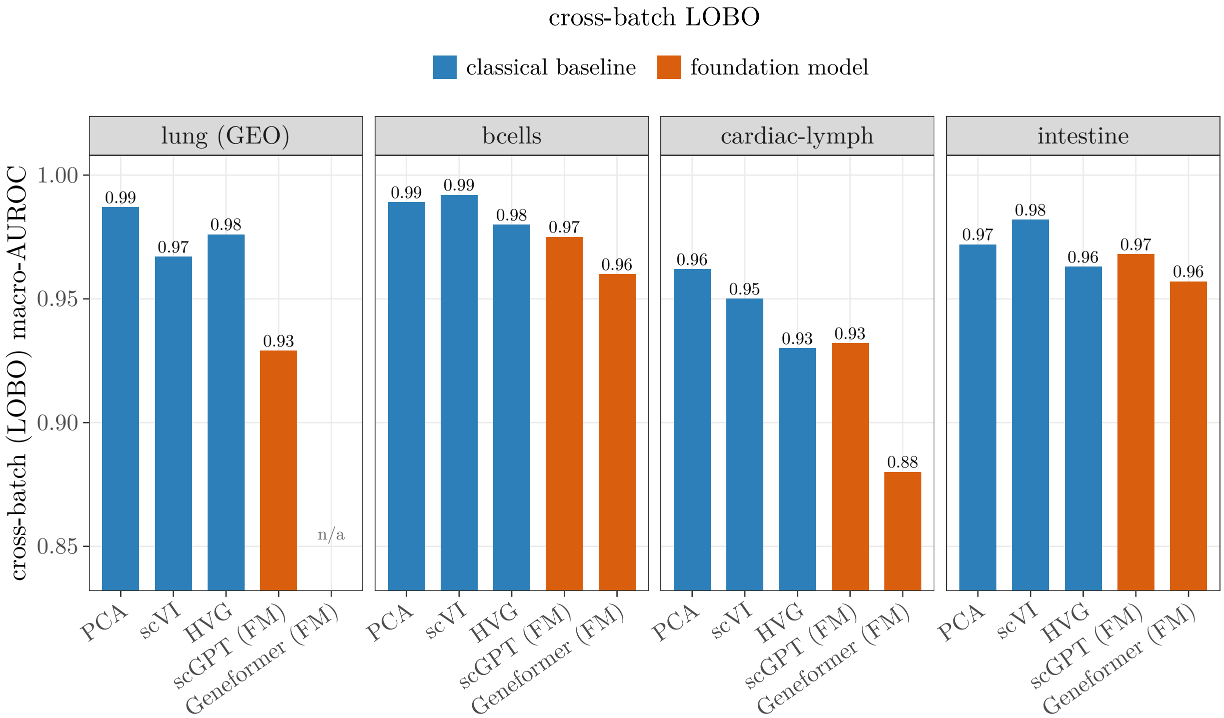


Figure 9: Self-computed (cluster J): on four leakage-controlled scRNA atlases, frozen zero-shot scFMs (scGPT, Geneformer) do not dominate classical baselines under leave-one-batch-out; PCA or scVI is best in every atlas.

Fig. 10). We do not report this as the result, because it is largely a metric artifact that a fair re-check (Fig. 2) corrects.

### 4.7 Self-computed scRNA suite: the fair re-check and parity (cluster J)

The fair re-check corrects that naive view on two fronts. First, a *non-linear* probe ($k$NN) removes the bias that a linear probe gives PCA (which lives in the same linear space the labels were derived from): under $k$NN, the best foundation model *ties or beats* the best simple baseline on all 11 vocabulary-matched human atlases (tie band $|\Delta\text{AUROC}| < 0.02$; e.g. stomach 0.97 vs. 0.93, breast-metastasis 0.91 vs. 0.85). Second, a *reference-free* structure metric—the fraction of highly-variable-gene expression variance explained by each method's unsupervised partition ($R^2_{\text{expr}}$, no labels used)—shows the FM and PCA partitions are essentially equal on the non-collapse atlases (tied-atlas means 0.163 vs. 0.172), and that the human cell-type labels themselves explain *less* expression variance than the PCA clustering in 20/20 atlases (and than *both* the FM and PCA clusterings in 11/20; e.g. stomach: labels 0.034 vs. PCA 0.157 vs. FM 0.138).

This is not an artifact of measuring variance in the 2000-HVG space (which could favor PCA): holding the partitions fixed and recomputing $R^2_{\text{expr}}$ at 1000, 2000, and 4000 HVGs and—decisively—over *all* genes (no HVG selection), the labels-below-PCA count is 20/20 in every case (the stricter below-*both*-clusterings count is likewise stable, 10–11/20 depending on which FM's clustering is compared), so the conclusion does not depend on the HVG bias the metric is sometimes charged with.

A complementary null control addresses the opposite objection—that $K$-means, by minimising within-cluster variance, is structurally favoured over any externally fixed labels on this metric:

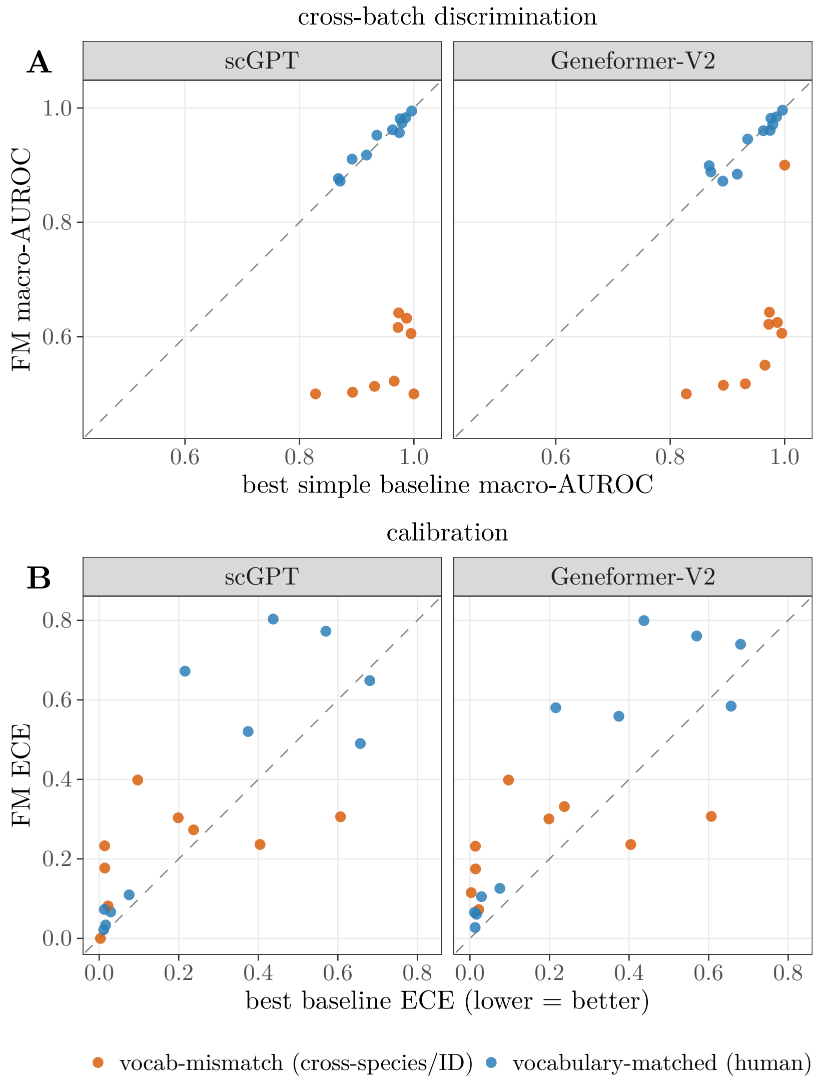


Figure 10: *The metric-artifact view* (cluster J): two zero-shot FMs (scGPT + Geneformer-V2) vs. the best simple baseline across 20 raw-count atlases (3 native + 17 unlocked by barcode label-transfer). **(A)** Cross-batch discrimination, scored by a *linear*-probe AUROC against the labels: the apparent baseline "win" is concentrated in the orange *vocab-mismatch* atlases (8 cross-species mouse-symbol + 1 ENSG-only, where the human-symbol FM tokenizer maps almost no genes); on the 11 blue vocabulary-matched atlases the points hug the diagonal, and the fair re-check (Fig. 2) removes the residual gap as circularity and probe bias. **(B)** The matched calibration view (FM ECE vs. best-baseline ECE, lower is better): the FMs sit mostly above the diagonal (worse-calibrated out-of-batch), the reliability counterpart that cluster K quantifies and partly repairs.

against a size-matched *random* partition the true labels explain ∼24× more HVG-expression variance (17/17 atlases; mean 0.110 vs. 0.005), so the labels are far from noise; the variance-optimised $K$-means partition does capture more (mean 0.171), yet the labels sit closer to that optimised ceiling than to the random floor (0.110, between 0.005 and 0.171). The reading is therefore not that the labels are biologically empty but that they encode a *coarser* partition than unsupervised clustering recovers.

Clustering the FM embedding instead of PCA cross-checks the vocabulary result at the same time: the labels fall below the FM clustering only on the matched atlases where the FM is functional (9/17), because on the cross-species/gene-ID atlases the human FM embedding is itself near-degenerate (its clustering's $R^2_{\text{expr}}\approx 0.004$) and cannot out-explain even the labels (Table A2).

The AUROC horse-race therefore measures "reproduce this particular (coarser) annotation cut" more than "capture the finer expression structure unsupervised clustering finds"; on that finer structure the FM and the baseline are tied.

That the labels under-explain expression variance is partly expected—$K$-means optimises within-cluster variance on the very HVGs scored—so we do not rest the circularity claim on the 20/20 count alone: its direct evidence is the probe asymmetry above (a linear probe, in the same linear space the labels were derived from, flatters PCA; the non-linear $k$NN probe removes the gap), and $R^2_{\text{expr}}$ corroborates it by showing the labels are a coarser cut than the structure unsupervised clustering—or the FM—recovers. A direct test corroborates it (on synthetic self-clustered labels, so illustrative rather than decisive—the load-bearing evidence stays the linear-vs-non-linear probe gap): generating labels by clustering each representation's own space and scoring every representation against *both* label sets, each representation scores higher against labels from its *own* space (per-atlas representation×label-source interaction +0.055, 95% CI [+0.039, +0.072], positive in 9/9 matched atlases), so the label-matching metric measures proximity to the label-generating method, not representation quality. The same coarseness inflates the label-matching metrics (ARI/scIB) used throughout the cited cluster-A literature.

What changes under this correction is the *location* of FM failure—and, on inspection, a second artifact as consequential as the circularity. Our earlier read attributed three "human" collapses to fine subtypes (T-cell, breast) and a louvain-labelled lung; but all three atlases in fact carry *mouse* gene symbols (`Xkr4`, `Gm`-genes, `Rik`-genes), so their collapse is the same cross-species token-map failure—not a representational one. Re-classifying all 20 atlases by their actual gene vocabulary leaves 11 vocabulary-matched human atlases (human symbols), 8 cross-species (mouse-symbol), and 1 gene-ID-mismatched atlas (ENSG-only, degenerate for the symbol tokenizer and trivially separable for PCA; excluded from the matched-parity equivalence set, though still shown as the 9th vocab-mismatch point in Fig. 2A).

On the 8 cross-species atlases the human-symbol FMs evaluated there (scGPT and Geneformer-V2, 104M/316M) drop to $k$NN≈0.53 (vs. PCA 0.81, which needs no gene map)—a disclosed, unfair tokenization handicap that the field's benchmarks rarely separate from representation. On the 11 vocabulary-matched atlases all five FM families are at parity with PCA on both the fair $k$NN probe (family means straddling PCA's 0.894 within a ≤0.024 spread) and the reference-free $R^2_{\text{expr}}$, each tying or beating PCA in 7–10 of 11 atlases (7 of 10 for UCE, which drops retina; per-family values in Table 4). The lone clean atlas where most FMs trail is bone marrow (0.83–0.88 vs. 0.91).

*Because "parity" is a null-equivalence claim, we test it as one rather than infer it from close means.* For each family we take its paired per-atlas difference from PCA ($k$NN-AUROC) and bootstrap the mean over the 11 atlases against a pre-stated equivalence margin of ±0.02 (the same tie band). No family differs significantly from PCA (Wilcoxon $p \geq 0.27$ for all six—no advantage *and* no deficit), and pooled across families the difference is +0.0025 (naive 95% CI [−0.006, +0.010]). Because the six arms share the same 11 atlases, those arm×atlas points are not independent; the correct interval is an atlas-cluster bootstrap (resampling atlases, the unit of independence), which roughly doubles the width to [−0.013, +0.020]. The equivalence is therefore positive but *marginal*: the cluster CI's upper edge (+0.0199) sits essentially *at* the ±0.02 margin—within the bootstrap's own Monte-Carlo error of it—so this is a boundary non-rejection. We read it as parity *at* the margin rather than a clean equivalence pass, and do not rest the claim on the TOST alone. One asymmetry matters: because the self-written loaders are pessimistic-for-the-FM (a better-tuned official pipeline can only *raise* an FM's score), the *no-deficit* half of this equivalence is firm—a real FM is at least as good as what we measure—while the *no-advantage* half is the conservative one, scoped to these faithful-but-unofficial loaders; scGPT and Geneformer-V2-316M certify parity on *both* sides without that caveat—their loaders are verified against official code (scGPT to three decimals; Geneformer-V2 to cosine 1.000, §3.6) and both pass the equivalence; for Geneformer-V2-104M the loader is equally verified, so its residual caveat is statistical power, not loader fidelity. Per arm the equivalence is certified for scGPT, Geneformer-V2-316M, and UCE, but for Geneformer-V2-104M, scFoundation, and CellPLM the per-arm CI exceeds $\pm 0.02$: with only 11 atlases we are underpowered to certify those three individually (their point estimates are within $\pm 0.011$ of PCA, but the interval is too wide to rule out a 0.02-scale effect).

We also avoid the "best-FM-per-atlas" summary, whose +0.024 apparent edge over PCA is a max-selection (winner's-curse) artifact; the per-family test above is the unbiased statement. The parity survives three robustness checks: it holds across probe neighbourhoods ($k \in \{5, 15, 30, 50\}$), is unchanged when the clustering is given twice as many clusters as labels ($R^2_{\text{expr}}$ gap 9/9 atlases at both $K$=NC and 2NC), and stays inside the tie band under two *non-distance* probes (multinomial logistic and a 128-unit MLP, both straddling zero)—so it is not an artifact of the $k$=15 choice, the cluster count, or the distance metric (if anything the MLP nudges the FM estimate slightly positive, i.e. $k$NN mildly under-rates it).

The corrected, defensible verdict: zero-shot scFMs neither beat nor trail simple baselines on vocabulary-matched coarse cell types—a parity, not the "baselines decisively win" our earlier single-metric read reported, which had averaged the cross-species handicap into the "human" mean and rewarded the method nearest the label-generating space. To test whether any residual shortfall is a *capacity* or *architecture* problem we added Geneformer-V2-316M (3× parameters, same loader) and three architecturally independent families via self-written/vendored loaders: scFoundation (Performer masked-autoencoder, autobin value encoding), CellPLM (flowformer encoder + GMVAE latent), and UCE (a protein-embedding gene tokenizer). Tripling parameters does not improve on the 104M sibling (316M mean 0.894 vs. 104M 0.905 across the 11 atlases), and the three further families—spanning radically different inductive biases—all land at the same parity (scFoundation 0.904, CellPLM 0.886, UCE 0.910; none significantly above or below PCA, as the equivalence test above shows). This is a direct, self-computed instance of "not governed by model scale or architecture family" (Table 4; see also [32] on the role of pre-training data scale and diversity).

### 4.8 Self-computed scRNA suite: the vocabulary-coverage artifact (cluster J)

A vocabulary dose–response makes this second artifact mechanistic, not binary. Rather than partition atlases into "matched" and "mismatched," we treat vocabulary coverage as a continuous dose: for every atlas×model pair we measure the fraction of the atlas's genes that map into that FM's gene vocabulary and plot it against the FM's zero-shot $k$NN-AUROC (Fig. 11). Apparent FM quality rises with coverage, but this is mostly a *cliff*, not a smooth ramp. The atlases split into a mismatch group that maps essentially no genes (the 8 cross-species atlases, and the ENSG atlas for scGPT; mapped fraction $\approx 0$) and a matched group that maps 32–94%, with *almost nothing in between*—the lone exception being the ENSG atlas under Geneformer-V2, which maps ∼24% and reads at parity (the same coverage effect, the other way). Across that full span the coverage–quality association is strong—pooling the three universally-run arms (scGPT, Geneformer-V2 104M/316M; 60 points) gives Spearman $\rho = 0.83$ (Pearson $r = 0.93$, $p < 10^{-26}$), reproduced in a reference-free metric (the FM clustering's $R^2_{\text{expr}}$, $\rho = 0.68$, $p < 10^{-8}$), so it is not a $k$NN-AUROC artifact—but it is dominated by the mismatch/matched gap: *within* the matched group alone the residual gradient is weak (Spearman $\rho \approx 0.34$ by coverage fraction for the universal-3 arms, 0.40 pooled across all families, and negligible, $\rho \approx 0$, by expression mass).

Table 4: Multi-FM check on the 11 vocabulary-matched human atlases (var_names = human gene symbols) under the fair $k$NN probe and a reference-free structure score ($R^2_{\mathrm{expr}}$, HVG-expression variance explained by each representation's own clustering). Five architecturally independent zero-shot scRNA FM families—rank-value (Geneformer), value-binning (scGPT), Performer masked-autoencoder (scFoundation), flowformer+GMVAE (CellPLM), and protein-embedding tokenizer (UCE)—span mean $k$NN 0.886–0.910 *around* PCA's 0.894: parity, with no family decisively beating or trailing it, while HVG alone is clearly worse. Tripling parameters within a family (Geneformer 104M→316M) buys nothing, and the spread across families ($\leq$0.024) is smaller than within-atlas noise. "ties/beats" counts atlases where the FM is within 0.02 of, or above, PCA. The 9 *vocab-mismatch* atlases are excluded here and analysed separately (Fig. 2): on the 8 cross-species (mouse-symbol) atlases the human-symbol FMs run there (scGPT, Geneformer-V2) drop to $k$NN$\approx$0.53 from a tokenization failure rather than a representational one, while the 1 ENSG-only atlas is the exception where a partial-coverage FM (Geneformer-V2, $\sim$24% of genes mapped) instead stays near parity.

| representation | mean $k$NN AUROC | mean $R^2_{\mathrm{expr}}$ | ties/beats PCA |
|---|---|---|---|
| best simple baseline (PCA) | 0.894 | 0.219 | 11/11 |
| scGPT | 0.894 | 0.205 | 10/11 |
| Geneformer-V2-104M | 0.905 | 0.219 | 9/11 |
| Geneformer-V2-316M | 0.894 | 0.225 | 9/11 |
| scFoundation | 0.904 | 0.215 | 8/11 |
| CellPLM | 0.886 | 0.168 | 7/11 |
| UCE[†] | 0.910 | 0.210 | 7/10 |
| HVG (2000 genes) | 0.750 | 0.232 | 0/11 |

[†]UCE applies its own min-count cell filter; one atlas (retina, 5 cells dropped) fell out of position-alignment and is omitted, so UCE is scored on 10/11.

The defensible reading is therefore a vocabulary *threshold* rather than a dose: atlases below it (tokenizer reads almost no genes) sit near chance (mean AUROC 0.53), those above plateau at PCA parity (mean 0.90 $\approx$ PCA's 0.894), and once the genes are legible more coverage buys little. The "FM failures" the field's cross-species and gene-ID atlases would record are thus a tokenization cliff—the representation is never tested when the vocabulary cannot read the genes—not a deficit that survives once it can.

*We confirm this causally, within a single atlas* (Fig. 11B): on three fixed matched atlases we keep only a fraction of the genes the tokenizer can read—renaming the rest so they map to nothing—and re-embed. Holding the atlas fixed removes the cross-atlas biology confound, and the tokenizer-free control is built in: PCA on the *identically* renamed matrix is unchanged (it never reads gene symbols, so renaming destroys no expression information), which isolates the effect as tokenization loss rather than information loss. The result is a clean *plateau-then-cliff*: $k$NN-AUROC stays at parity (0.92–0.97) from the full vocabulary down to $\sim$1–2k readable genes, then collapses, hitting exactly chance (0.500) at zero readable genes (Spearman frac-vs-AUROC 0.52–0.95 across the three). So vocabulary coverage *causes* the apparent quality: the FM needs only $\sim$its top-gene input budget worth of legible genes to match PCA, and the cross-species atlases (and the ENSG atlas for scGPT) sit at $\sim$0—below the cliff—which is exactly why they read as "FM failures" (the ENSG atlas is the exception: Geneformer-V2 maps $\sim$24% of it and reads at parity).

The cliff is *FM-general*, not specific to one tokenizer: repeating the ablation with a second, architecturally distinct family (scGPT, value-binning rather than rank-value) gives the same plateau-then-cliff on all three atlases (Spearman 0.73–0.98, parity down to $\sim$1–2k readable genes, chance at zero).

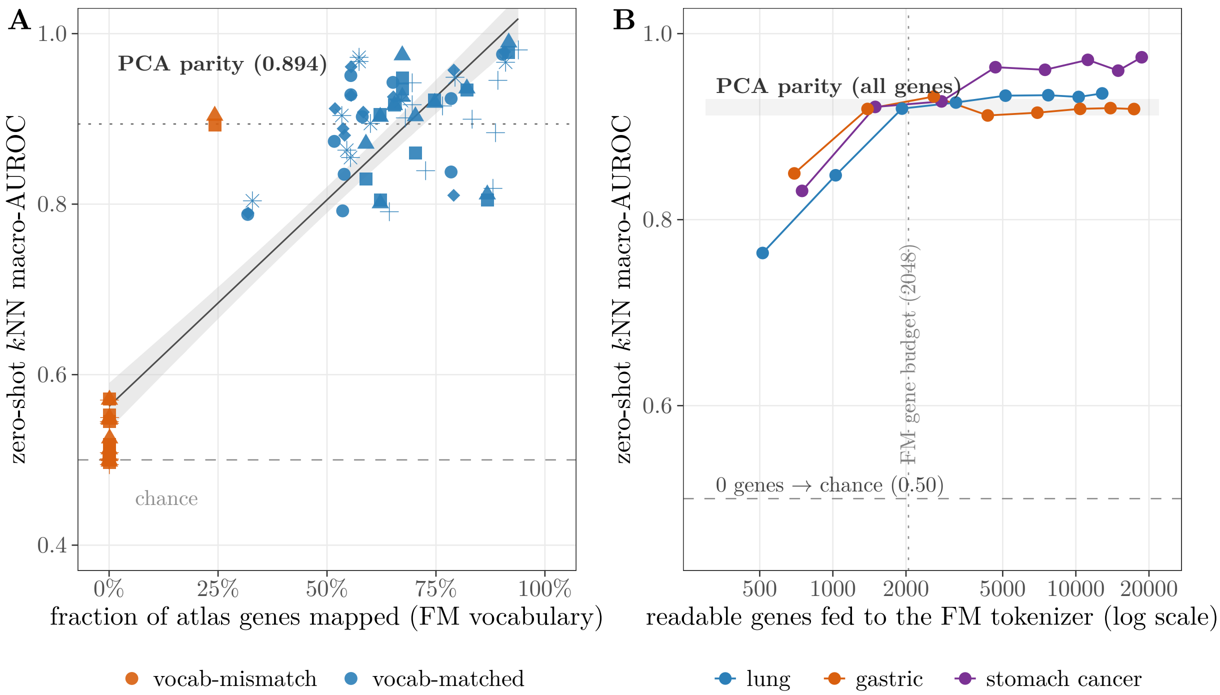


Figure 11: *The vocabulary artifact (cluster J, second artifact).* **(A)** *Dose-response:* each point is one atlas scored by one FM; $x$ is the fraction of the atlas's genes that map into that FM's gene vocabulary, $y$ is zero-shot $k$NN macro-AUROC. Apparent FM quality tracks coverage (universal-3 Spearman $\rho = 0.83$, Pearson $r = 0.93$), but mostly as a *threshold*: the two visible clusters—mismatch at $x \approx 0$ and matched at $x \in [0.32, 0.94]$, with almost nothing between (one ENSG point at $x \approx 0.24$ excepted)—bracket a near-empty gap, and within the matched cluster the residual gradient is weak ($\rho \approx 0.34$–$0.40$ by coverage fraction, $\approx 0$ by expression mass). **(B)** *Causal within-atlas ablation:* on three fixed *matched* atlases we keep only a fraction of the genes the tokenizer can read (renaming the rest so they map to nothing) and re-embed ($x$ = readable genes, log scale). Holding the atlas fixed removes cross-atlas biology; AUROC is flat at parity (grey band = full-gene PCA) down to ∼1–2k genes then cliffs to exactly 0.50 at zero readable genes. So the cross-species/gene-ID "FM failures" are a tokenization cliff (the FM needs only ∼its top-gene budget worth of legible genes to match PCA), not a representational deficit.

### 4.9 Self-computed scRNA suite: calibration and reliability (cluster K)

Frozen scGPT cross-batch *discrimination* transfers (LOBO macro-AUROC ∼0.95; shuffled-label control 0.48) but its *calibration* does not: cross-batch ECE is 0.17–0.23, and increasing read-out capacity (linear→MLP-256) uniformly worsens out-of-batch ECE in 4/4 atlases with no in-distribution gain (Fig. 12). Temperature scaling [42] closes only ∼1/3 of the gap (0.176→0.126). Split-conformal coverage is *unstable across batches*: the cross-batch coverage gap is positive in 3/3 non-lung atlases (0.012–0.075) and reaches 0.123 [0.115, 0.131] on the lung atlas under a genuine binomial count down-sample-and-re-embed depth ablation (the gap grows monotonically as depth drops, 0.123→0.169). This scGPT instability is genuine but, as the 24-atlas expansion below shows, sits at the *mild* end of a collapse that is larger for the classical baselines—so it is a real failure of exchangeability, not an FM-specific pathology. After deconfounding by log-depth, log-gene count, and batch (Frisch–Waugh–Lovell), confidence retains a residual benchmark-correctness signal (partial $r$ 0.25–0.40); we frame this as a cross-batch/exchangeability instability [17], not a leakage or biological-validity claim. This is the *self-computed scRNA half* of the modality contrast that the scATAC audit (cluster I) inverts, and the closest prior conformal-annotation work [43] targets neither a frozen FM nor a coverage-gap-under-shift measurement.

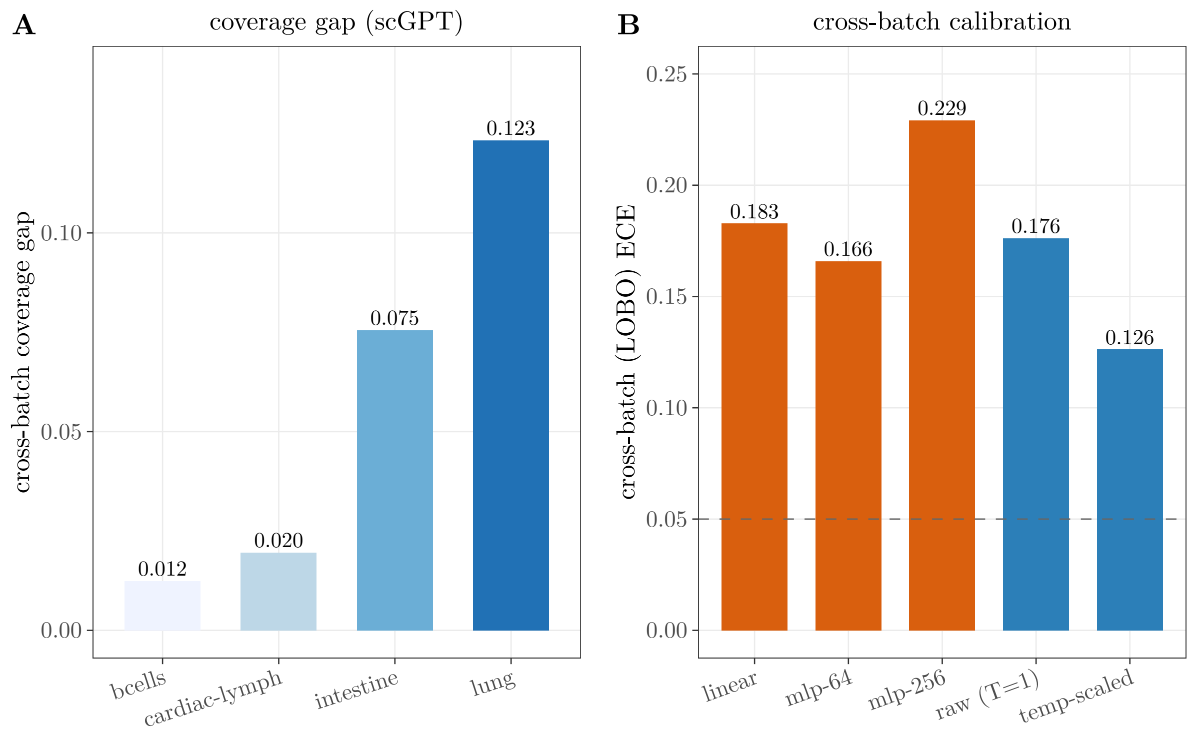


Figure 12: Self-computed (cluster K): frozen scGPT cross-batch conformal coverage is unstable (gap $> 0$ in all four atlases—0.012–0.075 in three, and 0.123 on lung under the depth ablation), and discrimination transfers while calibration does not—the highest-capacity readout (MLP-256) worsens ECE and temperature scaling closes only $\sim$1/3 of the gap.

*Expanding to 24 atlases reframes the finding.* Running the same split-conformal protocol across the 24-atlas panel and the full simple-baseline set shows the cross-batch coverage gap is *not* an FM-specific defect: it is large and pervasive for *simple* baselines too — mean gap +0.35 [0.21, 0.51] for PCA+logreg, +0.38 for random-forest, +0.37 for $k$NN, with a positive (under-coverage) gap in 20/24 atlases (max 0.92) (Fig. 14).

The frozen scGPT lung value (0.123) is in fact at the *milder* end of this distribution. The correct, more defensible claim is therefore that cross-batch conformal coverage collapse is a general exchangeability failure of single-cell representations under batch shift [17], afflicting classical baselines and foundation models alike (often the baselines more), rather than a calibration pathology unique to scFMs.

What remains FM-relevant is the negative: the foundation models do not *repair* this failure. The robust contribution is the *modality contrast* — this collapse is ubiquitous in scRNA cross-batch annotation (20/24 atlases, all methods) yet absent in scATAC coarse-type annotation under sample and disease shift (cluster I) — which localizes the problem to the scRNA exchangeability violation, not the model class.

The collapse is not fatal in practice: across the 24 atlases selective abstention recovers accuracy (accuracy at 80% coverage exceeds full-coverage accuracy for every baseline) and temperature scaling lowers—but does not erase—the cross-batch ECE (Fig. 13), so recalibration and a reject option are partial, deployable mitigations rather than a cure.

*Two robustness checks back this reading* (Fig. 15). First, the readout-capacity→out-of-batch-ECE effect is not scGPT-specific: it reproduces on a second FM family (Geneformer-V1-10M, pooled leave-one-batch-out), where the MLP-256 readout is again the worst-calibrated and discrimination stays flat—so over-capacity miscalibration is a property of the frozen-embedding readout, not of one model. Second, on the lung scGPT probe the confidence-based reject option is both effective (selective AURC 0.084 vs. 0.259 for random abstention) and *fair to rare cell types*: rare types are retained *more* than common ones at 80% coverage (0.90 vs. 0.79, i.e. not over-rejected), and the rare cells it does reject are disproportionately errors (rejected-rare accuracy 0.54 vs. retained-rare 0.91).

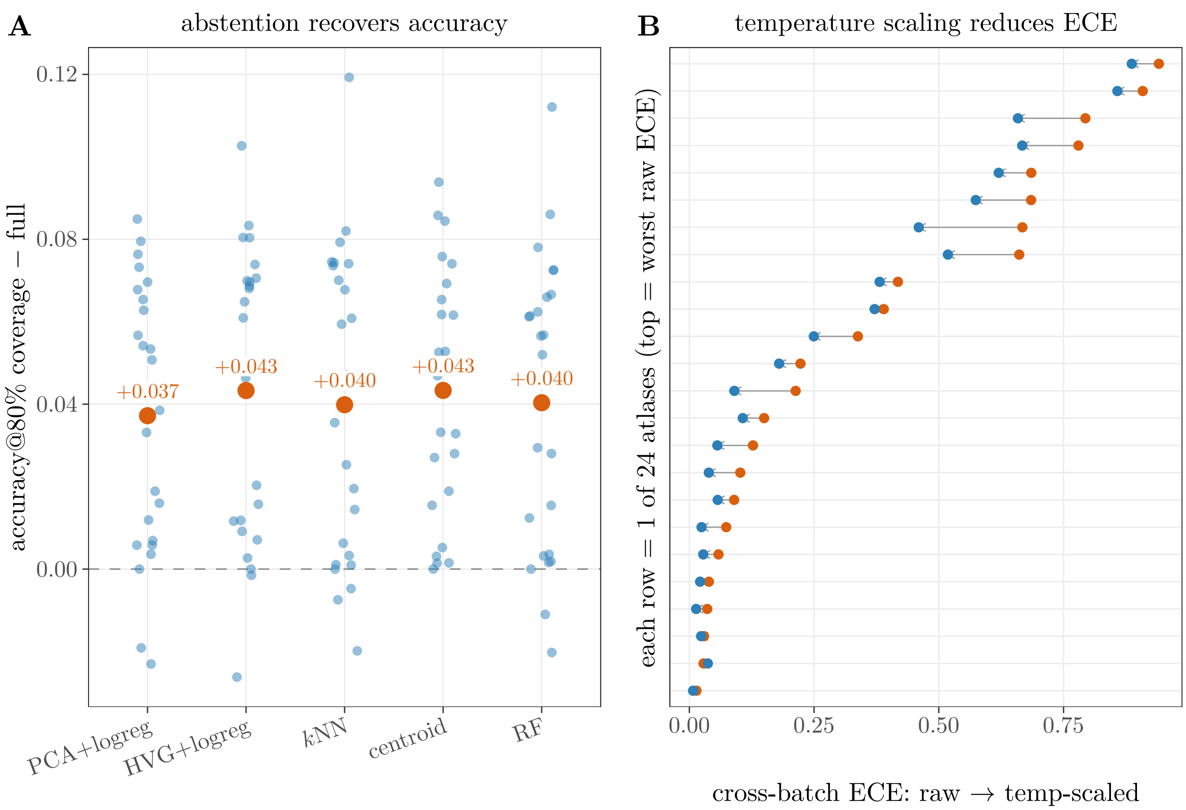


Figure 13: Self-computed scRNA reliability across 24 atlases—the *constructive* flip-side of cluster K. **(A)** selective abstention recovers accuracy (accuracy at 80% coverage exceeds full-coverage accuracy for every baseline). **(B)** temperature scaling lowers cross-batch ECE (orange→blue) in most atlases but does not eliminate it—so abstention/recalibration are partial remedies for the coverage collapse, not a cure.

*A batch-shift dose–response makes the "general exchangeability failure" claim quantitative.* If the collapse is genuinely driven by exchangeability violation rather than by the model class, its magnitude should scale with *how far* the held-out batch departs from the training batches. We measure that departure directly: in the same PCA space the coverage gap is computed in, we train a cross-validated classifier to predict whether a cell belongs to the held-out batch; its AUROC (0.5 = exchangeable, 1.0 = fully separable) is the exchangeability-violation strength. Across the

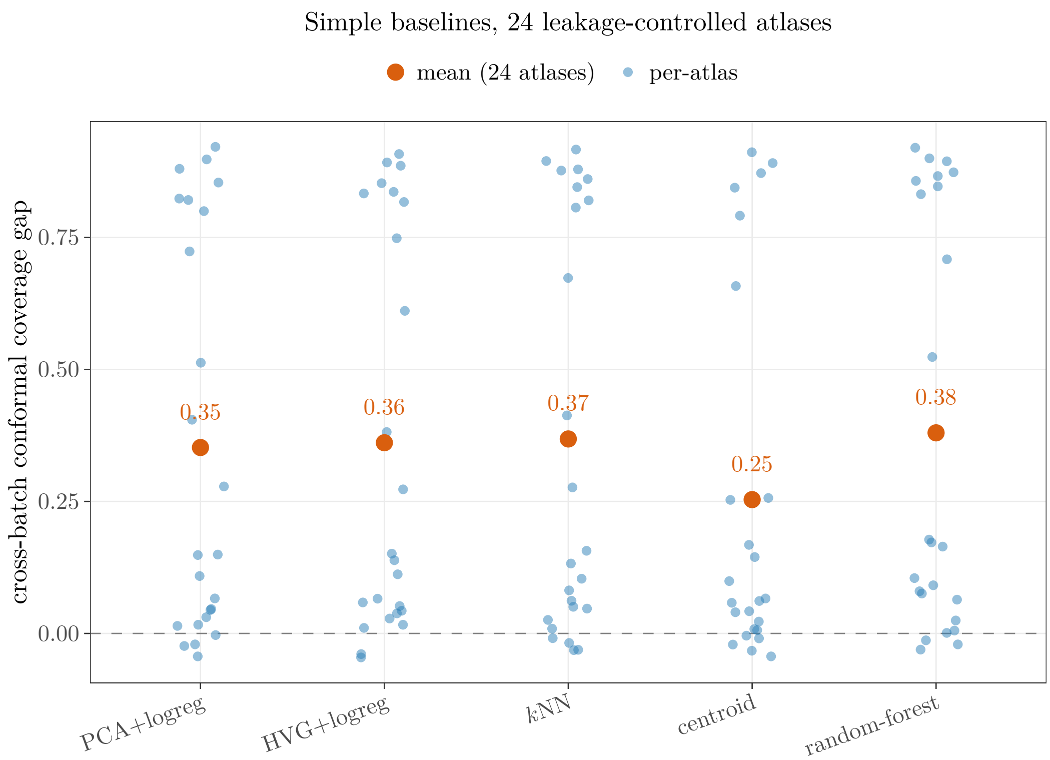


Figure 14: Expanded (cluster K reframe): across 24 leakage-controlled scRNA atlases, cross-batch conformal coverage collapse is large and pervasive for *simple* baselines too (each blue point is one atlas; the labelled orange point per method is the across-atlas mean; positive in 20/24), showing it is a general exchangeability failure under batch shift, not an FM-specific defect.

24 atlases the coverage gap is a graded function of this batch-shift strength—Spearman $\rho = 0.82$ (PCA+logreg; $p < 10^{-5}$), and the relationship survives every robustness check we applied (Fig. 7).

It is *method-general*: pooling the four PCA-based baselines (96 points) gives $\rho = 0.75$ ($p < 10^{-18}$), and an independent representation (HVG+logreg, with its own HVG-space shift measure) gives $\rho = 0.80$. It is not an artifact of a single degradation metric: a second reliability metric—the cross-batch *calibration* gap (test-batch ECE minus in-batch ECE)—tracks shift just as strongly ($\rho = 0.76$ across all representations). It is not an artifact of a single shift measure: a classifier-free alternative (the standardized centroid distance between the held-out and training batches) reproduces the trend ($\rho = 0.58$–$0.60$ for both degradation metrics).

And it is predominantly a failure of *exchangeability*, not of discriminative signal: at high shift the classifier still ranks held-out cells well (cross-batch discrimination AUROC 0.92, only 0.06 below in-batch) while coverage drops by 0.49—an $\sim 8\times$ larger effect—so the conformal sets miscover because the nonconformity-score distribution shifts, not because the task became unlearnable (discrimination does fall with shift too, $\rho = 0.83$, but $\sim 8\times$ smaller in magnitude than the coverage collapse).

*Foundation models collapse on the same curve.* Adding three FM families (scGPT, Geneformer-V2-104M, scFoundation) as the representation under the same protocol, the FMs—run on vocabulary-matched atlases that are high-shift in *both* the PCA and the FM embedding space (FM-space shift tracks PCA-space shift to within 0.011 across all 11, Pearson 0.97, so the embeddings neither create nor remove the batch separability; the only low/mid-shift atlases available are the mouse-symbol ones where the FM tokenizer fails, leaving no human atlas at

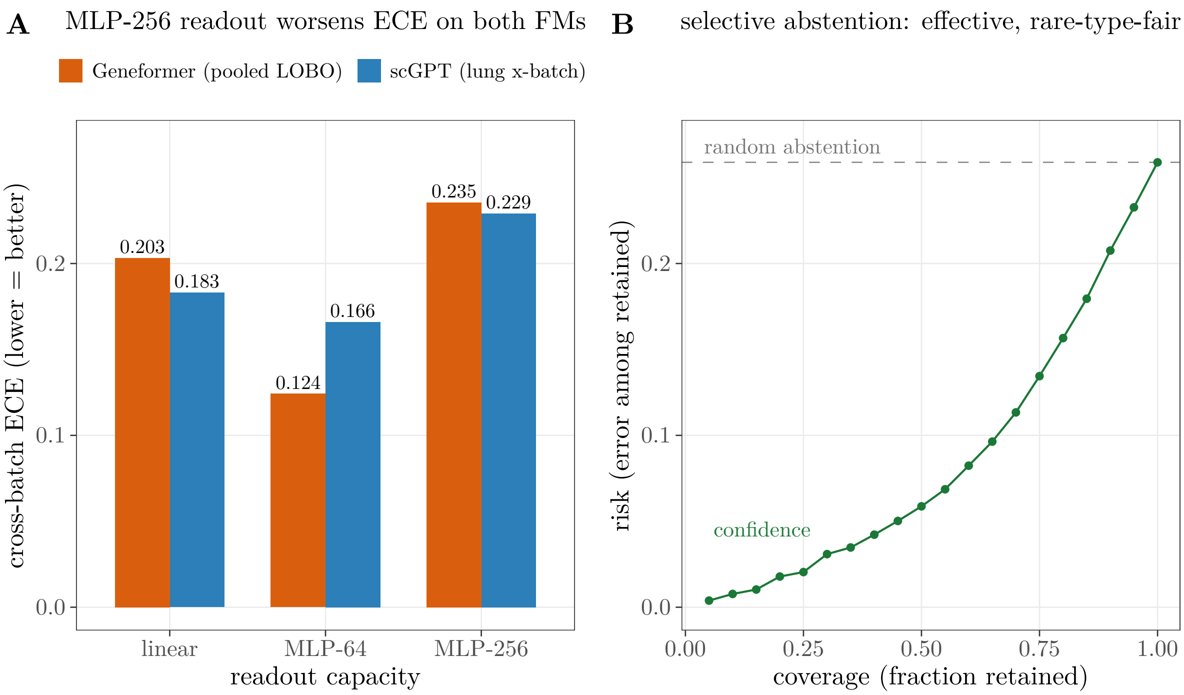


Figure 15: Cluster-K robustness checks. **(A)** The readout-capacity→out-of-batch-ECE effect is FM-general: the highest-capacity readout (MLP-256) is worst-calibrated for both scGPT (lung cross-batch) and a second family, Geneformer-V1-10M (pooled leave-one-batch-out over three atlases); MLP-64 is best for both, so it is an over-capacity effect, not a monotone one. **(B)** Selective abstention on the lung scGPT cross-batch probe: the confidence-based reject option (green) sits far below random abstention on the risk–coverage curve (AURC 0.084 vs. 0.259), and is *fair to rare cell types*—they are retained more than common types (0.90 vs. 0.79), and the rare cells rejected are disproportionately errors (rejected-rare accuracy 0.54 vs. kept-rare 0.91).

which to place a low-shift FM point)—reach only the *high-shift end* of the range (every FM point has shift $\geq 0.96$; Fig. 7A). There they collapse *by the same amount* as the classical baselines (mean coverage gap +0.48 vs. +0.49; ECE gap likewise), rather than repairing it. (Because the FM points span only this narrow high-shift band, we do not read a separate FM dose–response slope from them; the graded curve is the full-range classical sweep.)

Atlases whose held-out batch is near-exchangeable (shift AUROC $< 0.9$) show essentially no collapse (classical mean gap +0.03), while strongly non-exchangeable atlases collapse by an order of magnitude more (mean gap +0.5). The collapse therefore tracks the data regime, not the representation: every method and every model—classical or foundation—loses coverage in proportion to how badly the held-out batch breaks exchangeability, which is exactly what a foundation model would need to *repair* and does not.

*The same ruler makes the modality contrast quantitative.* Running the identical protocol on the scATAC atlas (cluster I; held-out *sample* as the batch, three representations—peak-LSI and two scATAC FMs, ChromFound and Atacformer) gives the opposite curve (Fig. 7B): the coverage gap is *flat* in batch-shift (Spearman $\rho = -0.24$, *n.s.*; ECE gap $\rho = -0.18$, *n.s.*), stays near zero throughout (mean $-0.005$, maximum $+0.08$ across 60 sample×representation points), and never collapses even for the most separable held-out sample (shift 0.96, gap $-0.07$). *The two slopes differ* *sharply:* regressing the coverage gap on batch-shift, the scRNA slope is +1.61 against −0.14 for scATAC (modality×shift interaction $t = -4.6$, $p = 1.6 \times 10^{-5}$; a label-permutation test on the slope difference gives $p = 3 \times 10^{-4}$). We read this as a *within-dataset descriptive* contrast, not a modality-level inference: the scATAC arm is a single atlas (GSE174367), so its unit of replication is one and the $p$-value indexes within-atlas sample variation rather than a general modality property—a modality-level claim would need $\geq 2$ scATAC datasets (§Limitations). Two things differ from scRNA: scATAC samples are also *more exchangeable* (shift tops out at 0.96, median 0.68, versus scRNA batches that routinely exceed 0.95), and—where the shift ranges overlap (0.85–0.96)—the scATAC gap stays at $\approx 0$ (mean −0.034) while the scRNA gap has begun to climb (mean +0.033). The contrast is therefore not an FM property but a *modality × regime* property: coverage collapse is specific to scRNA cross-batch annotation, and absent in scATAC coarse-type annotation under sample and disease shift, for classical representations and foundation models alike.

### 4.10 Self-computed scRNA suite: perturbation (cluster L)

On Norman-2019 [51] double perturbations, an additive linear prior (control $+\,\delta_A + \delta_B$) outperforms the deep GNN GEARS [30]. A *label-fair, full-gene* comparison (no oracle gene selection)—the test required before asserting GEARS is beaten—confirms it: additive beats GEARS on *all three* combo strata (both-singles-seen full-gene MSE 0.0016 vs. 0.0051, ∼3.2×; Fig. 16), so cluster L is *supported*, consistent with cluster B and the external linear-baseline result [26]. Separately, GEARS' own predicted uncertainty carries a genuine but modest selective-prediction signal that survives effect-magnitude deconfounding (partial Spearman 0.65; abstaining on the top-30% most-uncertain cells lowers retained MSE 0.0776→0.0734 vs. a shuffled-uncertainty control 0.0775).

## 5 Discussion

Three conclusions are robust across eleven task clusters (labels A–L, with E and F merged into one GRN cluster; summarised in Table 1), spanning external aggregation and self-computed re-derivation, and they are the three the abstract states. First, zero-shot scFMs provide no advantage—and no deficit—over simple/classical baselines on vocabulary-matched coarse cell types: they neither beat nor trail them on representation/annotation/integration and spatial niche identification, are real-but-metric-contingent for perturbation, and (newly) give no gain over their own input for a real scATAC FM. We state this as "no advantage" rather than "baselines decisively win" because much of the apparent baseline margin is not representational but two separable artifacts our fair re-check isolates. The first is a label-matching circularity: the field's standard scores (ARI, NMI, scIB, label-AUROC) are matched against cell-type labels derived from the same expression space the linear baselines occupy, and when we re-score with a non-linear probe and a label-free structure metric ($R^2_{\text{expr}}$) the FM and the baseline are tied, the labels proving a coarser partition than the clustering—explaining less expression variance than PCA in 20/20 atlases, and than both clusterings in 11/20 (Fig. 2)—so agreement-with-label ranks the annotation cut, not the finer expression structure. The second is a vocabulary-coverage threshold: genuine, label-independent FM shortfalls are confined to cross-species mouse-symbol and gene-ID-format atlases where the tokenizer reads almost no genes—a tokenization handicap, not a representational failure. Three mechanistic depth probes (Table 5) turn each of the paper's claims from a binary or qualitative assertion into a measured dose–response.

Second, the much-warned cross-batch calibration and conformal-coverage collapse is a *general* exchangeability failure of the scRNA data regime, not an FM-specific defect: it afflicts classical baselines as much as frozen embeddings (often the baselines more), grows with how badly the held-out batch breaks exchangeability (Fig. 7), and is *absent* in scATAC under sample and disease shift—so what a foundation model would need to repair, and does not, is a property of the modality

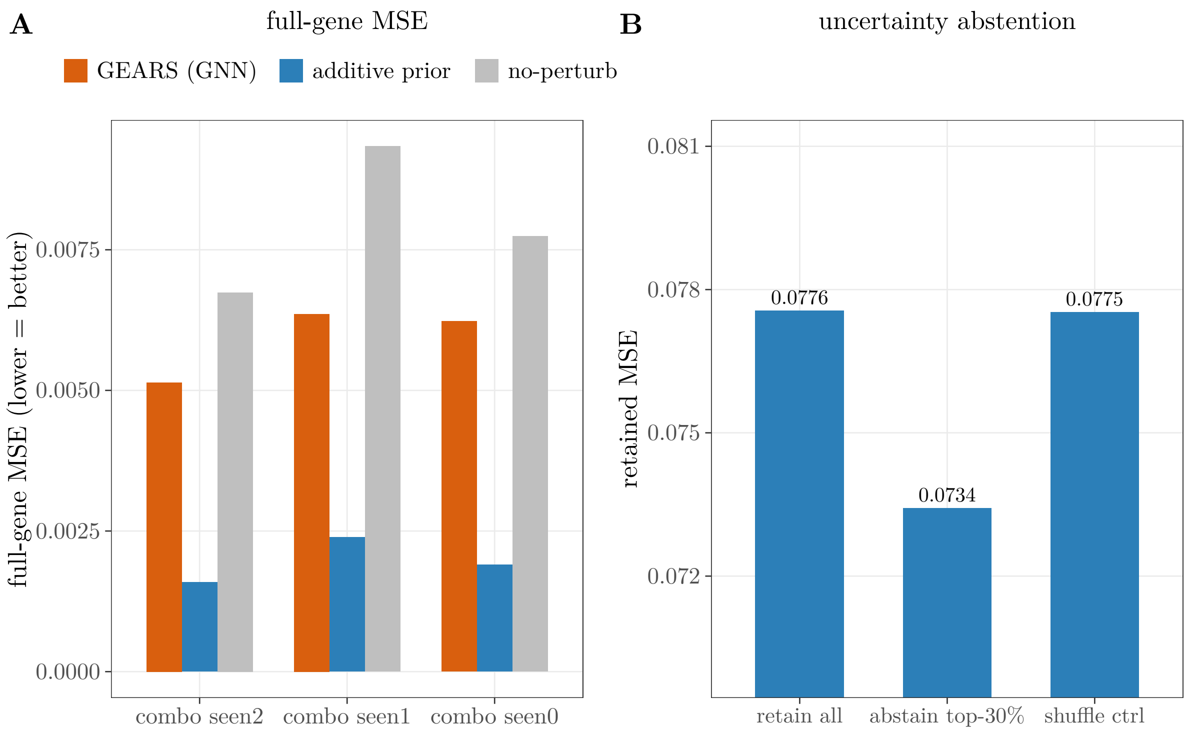


Figure 16: Cluster L (perturbation). **(A)** a label-fair, full-gene comparison (no oracle gene selection) on Norman-2019—the additive prior beats the deep GNN GEARS on all three combo strata (∼3× lower MSE; 2.7–3.3×), upgrading cluster L from exploratory to supported. **(B)** GEARS' own predicted uncertainty gives a genuine but modest selective-prediction gain (abstaining on the top-30% most-uncertain cells lowers retained MSE; partial Spearman 0.65 after effect-magnitude deconfounding). The panel-B $y$-axis is truncated (bar labels give absolute MSE); the reduction is ∼5%, small in absolute terms.

and regime, not the model class. This finding carries the paper's constructive spine: reliability is recoverable where the data are exchangeable (scATAC coverage holds for every representation, Table 3) and partially repairable where they are not (selective abstention and recalibration claw back accuracy under scRNA batch shift, Fig. 13); and even where pretraining *does* appear to help—FM batch mixing—it is bought at a cell-type-purity cost that a purpose-built classical method (scVI) avoids while dominating the mixing/purity trade-off (Fig. 8).

Third, which method "wins" (when one does) is governed by task and evaluation choice—not by model scale or architecture—most visibly in perturbation (metric family), scATAC calibration (dimensionality), and the probe/label circularity above. We test the scale claim directly: tripling a model's parameters (Geneformer 104M→316M) and swapping in three architecturally independent families (scFoundation, CellPLM, UCE) leave the result unchanged (Table 4). The value of pretraining, where it appears, is in fine-tuned or supervised-downstream use, not zero-shot representation. The through-line across all three conclusions is that what determines whether a frozen embedding is trustworthy is the task, the metric, and data exchangeability—not model scale.

The backbone is our own in-house experiments—the self-computed scRNA reliability suite (five FM families, up to 24 atlases), the first spatial-FM niche head-to-head, and the first scATAC calibration audit—with the external meta-analysis as supporting context. The headline is methodological: a fair, reference-free re-scoring shows the standard label-matching leaderboards do not adjudicate what they claim.

We state these findings provisionally. They are one practitioner's results on tractable subsets with self-written loaders; the circularity argument rests on a reference-free metric that itself carries a mild expression-space bias (bounded with a size-matched random-partition null: the labels still explain $\sim$24$\times$ the null variance, so they are coarse, not empty), the FM arms are pessimistic lower bounds, and several clusters rest on few studies. So we do not conclude that foundation models are settled to be useless, nor that the baselines are vindicated—only that the field's current rules cannot adjudicate the question, and that a fairer, reference-free test is now on the table. Each claim is falsifiable and we invite refutation.

Table 5: The three mechanistic depth probes: each converts a binary or qualitative claim into a measured dose–response, and each is computed with the method-class the claim concerns (foundation models) on the curve. "Result" gives the headline; full statistics are in §3.6 and the cited figures.

| depth probe | scientific question | driver ($x$) | outcome ($y$) | result |
|---|---|---|---|---|
| #1 vocabulary (scRNA repr.) | apparent FM failure: representational or tokenization? | mapped-gene fraction | $k$NN-AUROC | *threshold* ($\rho$=0.83); causal within-atlas ablation is flat at parity then cliffs to chance below $\sim$1–2k genes; FM-general (scGPT & Geneformer) |
| #2 batch-shift (scRNA reliab.) | coverage collapse: FM defect or general? | held-out-batch separability | coverage gap (& ECE gap) | *general* ($\rho$=0.82); method- and model-general (FMs on the same curve); predominantly exchangeability, not task difficulty; scATAC flat ($\rho$=−0.24) |
| #3 spatial context (niche-ID) | residual gap: representational or a missing operation? | spatial smoothing $k$ | niche $k$NN-AUROC / F1 | *aggregation axis*: spatial-smoothing per-cell PCA overtakes all three spatial FMs (F1 $\rho$=0.98); the FMs simply are not given the dose |

**Implications**

For practitioners, three practical guidelines follow. (1) *Choose representations by task, not by pedigree*: on zero-shot coarse-type annotation, integration, and niche identification, a cheap classical pipeline (PCA, scVI, neighbourhood composition, spatial smoothing) is at least as good as a foundation model and far cheaper to run, so a foundation model should be adopted only where a concrete downstream gain—typically fine-tuned or supervised—has been demonstrated. (2) *Do not trust a single label-matching score*: because cell-type labels are themselves derived from the expression space, ARI/NMI/scIB/label-AUROC can rank the annotation pipeline rather than the biology; a non-linear probe plus a reference-free structure metric should accompany any such leaderboard, and "no difference" should be argued with an equivalence test, not an absent $p$-value. (3) *Treat cross-batch uncertainty as unreliable by default*: conformal coverage collapses with batch shift for classical and foundation representations alike, so a reject option (selective abstention) and recalibration are necessary deployable mitigations, and scATAC's robustness shows the failure is a property of the scRNA cross-batch regime rather than of the model.

Methodologically, the contribution we most want carried forward is the move from binary verdicts to *dose–response with the criticized method class on the curve*: tying an apparent failure to a measured, manipulable driver (vocabulary coverage, batch-shift strength, spatial-aggregation dose) converts a leaderboard rank into a mechanism, and—as the vocabulary ablation shows—can flip a "representational deficit" into a removable, causal artifact.

**Generality beyond genomics.** Nothing in the protocol is biological. It takes as input a frozen encoder [1, 2], a set of examples with labels of contested provenance, and a covariate along which the deployment distribution shifts—and returns a fair-evaluation verdict paired with a reliability audit. The three confounds it targets recur wherever pretrained representations are ranked: label circularity arises whenever "ground truth" is itself derived in, or downstream of, the representation space being graded (weakly-supervised vision labels, LLM-judged text, pseudo-labelled speech); tokenizer or vocabulary coverage is a measured variable for any model with a fixed input alphabet facing out-of-vocabulary symbols or cross-domain inputs; and the exchangeability that underwrites split-conformal coverage [17, 40, 41] and the calibration that temperature scaling repairs [42] break under any covariate shift, not just batch and donor effects. Each protocol move is correspondingly generic—a non-linear probe versus a linear comparator, a reference-free structure metric, an equivalence test in place of an absent $p$-value, and a dose–response that puts the criticized method on the curve against its own confound. Porting the instrument to a new domain requires only naming the dose (the shift covariate) and a label-independent structure metric appropriate to the modality; the leakage control, conformal machinery, and equivalence testing transfer unchanged. The claim we defend is therefore methodological: a leaderboard should certify that it measured a representation and not its own rules, and that certificate is domain-independent even though the stress test reported here is not.

### Limitations

Scope is strictly zero-shot. The pooled corpus is dominated in raw count by one study (mitigated by equal-weight estimation), cluster B rests on seven studies (several small), and—assembled by a literature search for FM-vs-baseline benchmarks—it is subject to selection and publication bias, the "simple beats FM" literature possibly over-represented; we ran no formal publication-bias diagnostic, a further reason the meta-analysis is supporting context, not an unbiased population estimate.

The self-computed *generality* is bounded by dataset count: the spatial-niche, scATAC, and perturbation arms each rest on a *single* dataset (one CosMx lymph node, GSE174367, Norman-2019)—mechanism demonstrations, not multi-dataset establishments—while only the scRNA suite spans many atlases (up to 24). The primary scATAC FM (ChromFound) ran on a disclosed top-2048-peak / 20k-cell subset (a plausible but unverified lower bound; no peak sweep; Atacformer used the 890k-region overlap on the same 20k-cell subset). The matched-vocabulary equivalence rests on 11 atlases and certifies for only 3 of 6 arms (the others consistent with, but underpowered for, parity); its matched/mismatch split was revised mid-analysis after a gene-naming audit, so we report both verdicts—a split sensitivity shows the pre-audit set (mis-filing three mouse-symbol atlases as human) turns parity into an apparent $-0.041$ deficit, i.e. the revision removed an anti-FM artifact on an objective criterion rather than engineering the parity. The 20-atlas FM arm relies on barcode label-transfer onto raw-count twins ($\geq$50% overlap), and the expanded reliability/simple-baseline panels omit scVI.

Several "ground-truth" labels are clustering-derived and thus partially circular for PCA; we correct with a non-linear probe and a reference-free $R^2_{\text{expr}}$, whose HVG-space bias we bound directly (the "labels < clustering" count is unchanged at 1000–4000 HVGs and over all genes). Reflexively, this re-check is itself a set of evaluation choices—a $k$NN probe, a $\pm 0.02$ margin, 11 atlases—so by our own third finding it cannot be the uniquely correct arbiter; we stress-test those choices (probe family and neighbourhood, cluster count, an atlas-cluster bootstrap, a direct circularity test) rather than assert them, and offer the re-check as falsifiable, not final.

Finally, the FM loaders make recipe assumptions and are faithful-but-not-official lower bounds—except scGPT and Geneformer-V2, whose loaders reproduce the official embeddings exactly (§3.6), while scFoundation, CellPLM and UCE run the authors' *own* code, so the five-family parity rests on others' code rather than our re-implementation. Integration-metric critiques (cluster D) caveat the scIB-based inputs to cluster A, and cluster L's additive-vs-GEARS result now rests on a label-fair full-gene comparison (the earlier oracle-gene version superseded), with the second-FM (Geneformer) reproduction qualified, as noted (§3.6).

## Data and code availability

All analysis code, the normalized comparison tables, the self-computed audit outputs, and the figure-generation scripts are archived on Zenodo with a versioned DOI (10.5281/zenodo.21071826), organized by cluster, and the curated analysis code and a reusable specification of the evaluation protocol are maintained at github.com/PeterPonyu/frozen-fm-eval; each stage—corpus pooling, the fair-probe re-analysis, the vocabulary / batch-shift / spatial dose–response analyses, the scATAC and spatial audits, the 24-atlas expansion, and the foundation-model embedders—is a standalone, deterministic script with fixed random seeds. External per-comparison tables are reused from each cited study's public release and are not redistributed (metric values only, ∼0.3 GB). Primary data are public: the snATAC atlas of Morabito *et al.* [50] (GEO accession GSE174367), the manually annotated single-cell CosMx (human follicular lymphoid hyperplasia) lymph-node niche dataset from the spatial niche-identification benchmark of Wang *et al.* [47] (file `lymph_node_niche_annotated.h5ad`, `https://github.com/WYXNICK/spatial-niche-benchmark`), and the Norman *et al.* 2019 Perturb-seq data (GEO accession GSE133344) [51], used as the GEARS-distributed preprocessed Norman-2019 AnnData; all foundation-model weights are obtained from each model's public repository. The 24 leakage-controlled scRNA atlases underlying clusters J and K (short-coded in Table A2) were assembled from public GEO/CELLxGENE releases; the complete short-code-to-accession-and-publication mapping for all 24 atlases and the 95-atlas inventory is provided in the released Zenodo archive. The four primary leave-one-batch-out scRNA atlases (Figs. 9–12; 4–26 donors) are likewise public: the lung atlas is GEO accession GSE130148, and the per-atlas accessions and source publications for the remaining primary atlases (B-cells, cardiac-lymph, intestine) are provided, together with those of the full suite, in the released atlas manifest (`atlas_manifest.json`) on Zenodo. Analyses run in a single GPU environment (PyTorch, scanpy, scikit-learn, and two conformal-prediction libraries [45, 46]); foundation-model inference uses an isolated environment.

## Author contributions

The analysis, audits, software, figures, and manuscript were produced as a single working effort. Independent adversarial verification was run as a separate pass and corrected an initial scATAC-FM overclaim into the matched-control statement reported here.

## Scope and ethics

All claims are restricted to the zero-shot / out-of-the-box regime and to benchmark reliability; no clinical, diagnostic, or biological-discovery claims are made. Fine-tuned or supervised-downstream foundation models are out of scope and can reverse these results. Disease-state data (Control/AD) is used only as a domain-shift axis for conformal coverage, not for any clinical inference.

## Competing interests

The author declares no competing interests.

## Use of AI

AI-based tools (Claude Code, Anthropic; Claude Opus and Claude Sonnet models; `https://claude.ai/code`) assisted analysis-code implementation and refactoring, manuscript language preparation, and computational verification. All AI-assisted code and text were reviewed, tested, and verified by the author, who takes full responsibility for the content; no AI tool is credited as an author.

## Abbreviations

scFM/FM, (single-cell) foundation model; scRNA, single-cell RNA-seq; scATAC, single-cell ATAC-seq; snATAC, single-nucleus ATAC-seq; PCA, principal-component analysis; HVG, highly variable genes; scVI, single-cell variational inference; VAE, variational autoencoder; GMVAE, Gaussian-mixture variational autoencoder; LSI, latent semantic indexing (TF-IDF → SVD); SVD, singular-value decomposition; TF-IDF, term-frequency inverse-document-frequency; *k*NN, *k*-nearest-neighbours; TOST, two one-sided tests (equivalence test); kBET, *k*-nearest-neighbour batch-effect test; GNN, graph neural network; MLP, multi-layer perceptron; DL, deep learning; LOBO, leave-one-batch-out; CV, cross-validation; EW, equal-weight (per study); CI, confidence interval; AUROC, area under the ROC curve; ARI, adjusted Rand index; NMI, normalised mutual information; scIB, single-cell integration benchmarking metrics; ECE, expected calibration error; LAC, least-ambiguous-set conformal prediction; AURC, area under the risk–coverage curve; FWL, Frisch–Waugh–Lovell (partial-correlation deconfounding); $R^2_{\text{expr}}$ / $R^2_{\text{compo}}$, fraction of single-cell-expression / neighbourhood-composition variance explained by a partition (reference-free, no labels); GRN, gene-regulatory network; UQ, uncertainty quantification; GEO, Gene Expression Omnibus; AD, Alzheimer's disease.

## Appendix: Contributing studies

Table A1 lists the studies that contributed usable per-comparison numbers (clusters A and B) plus the principal sources for the reliability/auxiliary clusters. The full catalogue of 37 entries (including infrastructure and counter-evidence), with an extractability manifest, is provided in the released Zenodo archive.

Table A1: Studies contributing to the meta-analysis, by cluster.

| Cluster | Study | Role |
|---|---|---|
| A | Kedzierska *et al.* 2025 [15] | zero-shot embedding eval |
| A | Boiarsky *et al.* 2024 [19] | scRNA FM deep dive |
| A | Wu *et al.* 2025 [22] | biology-driven post-release holdout |
| A | Atti & Subramaniam 2025 [21] | zero-shot embedding eval (A4) |
| A | Bendidi *et al.* 2024 [20] | "one PCA rules them all" |
| B | scPerturBench (Wei 2026) [28] | backbone perturbation corpus |
| B | Ahlmann-Eltze *et al.* 2025 [26] | linear-baseline perturbation |
| B | Csendes *et al.* 2025 [25] | post-perturbation RNA-seq |
| B | PertEval-scFM (Wenteler 2025) [27] | FM perturbation benchmark |
| B | Kernfeld *et al.* 2025 [24] | expression-forecasting benchmark |
| B | Wong *et al.* (Pfizer) 2025 [23] | simple controls vs. DL/FM |
| B | Viñas *et al.* (Systema) 2026 [29] | systematic-variation eval |
| E/F | Kendiukhov 2026 [31] | FM attention vs. GRN |
| E/F | scRegNet (Kommu 2025) [52] | FM-embedding + GNN GRN |
| G | veloBench (Luo 2026) [53] | RNA-velocity benchmark |
| D | RBET [39], scIB-E [37], scIB [34] | integration-metric critiques |
| C | popV [35], HCE [36], Theunissen [44] | consensus/calibrated UQ |
| H | self-computed (this work) | spatial FMs vs. niche baseline |
| I | self-computed (this work) | scATAC calibration; ChromFound [11], Atacformer [12] |
| J | self-computed (this work) | scRNA LOBO: scGPT/Geneformer vs. PCA/scVI [13] |
| K | self-computed (this work) | scRNA cross-batch conformal coverage; temp scaling [42] |
| L | self-computed (this work) | GEARS [30] vs. additive on Norman-2019 [51] |

Table A2: *Null-partition control for the label-vs-clustering* $R^2_{\mathrm{expr}}$ *comparison* (cluster J; held-out batch of each matched/mismatch raw-count atlas). Columns are the fraction of held-out HVG-expression variance explained by: a size-matched *random* partition (permuted labels), the true cell-type *labels*, the *PCA K*-means clustering ($K$=#labels), and the best-available *FM K*-means clustering. The labels beat the random null in 17/17 atlases (so they are not noise) yet fall below the variance-optimised PCA clustering in 17/17 (the cluster-J finding, here with a null floor). They fall below the *FM* clustering in only 9/17: on the cross-species/gene-ID atlases the human FM embedding is itself near-degenerate (FM-clustering $R^2_{\mathrm{expr}} \approx 0$, lower rows), the same tokenization failure as Fig. 11, read through the clustering. This control covers the 17 of the 20 atlases for which a size-matched random partition was computed; two of the three omitted atlases are among the 11/20 where labels fall below *both* clusterings, so the 9/17 here corresponds to that 11/20.

| atlas | #labels | random | labels | PCA-clust | FM-clust |
|---|---|---|---|---|---|
| ad_hm | 15 | 0.001 | 0.226 | 0.306 | 0.285 |
| breast_hm | 18 | 0.004 | 0.192 | 0.235 | 0.026 |
| ca_cancer | 17 | 0.003 | 0.190 | 0.266 | 0.252 |
| gastric | 20 | 0.006 | 0.174 | 0.214 | 0.201 |
| hesc_hspc_cd8 | 15 | 0.003 | 0.169 | 0.196 | 0.014 |
| astrocytes_sci | 20 | 0.007 | 0.164 | 0.211 | 0.040 |
| lung_adre | 22 | 0.011 | 0.138 | 0.209 | 0.193 |
| tcell_cancer | 16 | 0.005 | 0.132 | 0.175 | 0.026 |
| bm_all | 13 | 0.006 | 0.118 | 0.189 | 0.179 |
| progastin | 21 | 0.003 | 0.064 | 0.122 | 0.004 |
| urine | 24 | 0.003 | 0.062 | 0.110 | 0.005 |
| lsk_batch | 11 | 0.002 | 0.056 | 0.076 | 0.003 |
| breast_hm2 | 18 | 0.007 | 0.055 | 0.212 | 0.195 |
| lps_mm | 4 | 0.001 | 0.054 | 0.071 | 0.000 |
| stomach_cancer | 17 | 0.005 | 0.034 | 0.157 | 0.138 |
| aml_pbmc | 19 | 0.005 | 0.029 | 0.063 | 0.053 |
| breast_metastasis | 26 | 0.006 | 0.018 | 0.088 | 0.079 |
| *mean* | | 0.005 | 0.110 | 0.171 | 0.099 |

## References


[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. doi: 10.48550/arXiv.1706.03762. arXiv:1706.03762.

[2] R. Bommasani, D. A. Hudson, E. Adeli, et al. On the opportunities and risks of foundation models, 2021. Stanford CRFM; arXiv:2108.07258.

[3] A. Regev, S. A. Teichmann, E. S. Lander, et al. The Human Cell Atlas. *eLife*, 6:e27041, 2017. doi: 10.7554/eLife.27041.

[4] The Tabula Sapiens Consortium. The Tabula Sapiens: a multiple-organ, single-cell transcriptomic atlas of humans. *Science*, 376(6594):eabl4896, 2022. doi: 10.1126/science.abl4896.

[5] H. Cui, C. Wang, H. Maan, K. Pang, F. Luo, N. Duan, and B. Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21, 2024. doi: 10.1038/s41592-024-02201-0.

[6] C. V. Theodoris, others, and P. T. Ellinor. Transfer learning enables predictions in network biology. *Nature*, 618:616–624, 2023. doi: 10.1038/s41586-023-06139-9.

[7] M. Hao, others, and L. Song. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 21, 2024. doi: 10.1038/s41592-024-02305-7.

[8] Y. Rosen, Y. Roohani, A. Agrawal, L. Samotorcan, Tabula Sapiens Consortium, S. R. Quake, and J. Leskovec. Universal cell embeddings: a foundation model for cell biology, 2023. bioRxiv.

[9] H. Wen, W. Tang, X. Dai, J. Ding, W. Jin, Y. Xie, and J. Tang. CellPLM: pre-training of cell language model beyond single cells. In *ICLR*, 2024. doi: 10.1101/2023.10.03.560734. bioRxiv 2023.10.03.560734.

[10] A. Tejada-Lapuerta, others, and F. J. Theis. Nicheformer: a foundation model for single-cell and spatial omics. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02814-z.

[11] Y. Jiao et al. ChromFound: towards a universal foundation model for single-cell chromatin accessibility data, 2025. arXiv:2505.12638.

[12] N. J. LeRoy, G. Zheng, O. Khoroshevskyi, D. R. Campbell, A. Zhang, and N. C. Sheffield. Atacformer: a transformer-based foundation model for analysis and interpretation of ATAC-seq data, 2025. URL https://huggingface.co/databio/atacformer-base-hg38. bioRxiv; Hugging Face model databio/atacformer-base-hg38; gtars.

[13] R. Lopez, J. Regier, M. B. Cole, M. I. Jordan, and N. Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15:1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.

[14] I. Korsunsky, N. Millard, J. Fan, K. Slowikowski, F. Zhang, K. Wei, Y. Baglaenko, M. Brenner, P.-R. Loh, and S. Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0.

[15] K. Z. Kedzierska, L. Crawford, A. P. Amini, and A. X. Lu. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26:101, 2025. doi: 10.1186/s13059-025-03574-x.

[16] F. A. Wolf, P. Angerer, and F. J. Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19:15, 2018. doi: 10.1186/s13059-017-1382-0.

[17] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2):816–845, 2023. doi: 10.1214/23-AOS2276.

[18] J. Wu et al. EpiFoundation: a foundation model for single-cell ATAC-seq via peak-to-gene alignment, 2025. bioRxiv.

[19] R. Boiarsky, N. M. Singh, A. Buendia, G. Getz, and D. Sontag. Deeper evaluation of a single-cell foundation model. *Nature Machine Intelligence*, 2024. doi: 10.1038/s42256-024-00949-w.

[20] I. Bendidi et al. Benchmarking transcriptomics foundation models for perturbation analysis: one PCA still rules them all, 2024. arXiv:2410.13956.

[21] S. Atti and S. Subramaniam. Fundamental limitations of foundation models in single-cell transcriptomics, 2025. bioRxiv.

[22] J. Wu et al. Biology-driven insights into the power of single-cell foundation models. *Genome Biology*, 26:334, 2025. doi: 10.1186/s13059-025-03781-6.

[23] D. R. Wong, A. S. Hill, and R. Moccia. Simple controls exceed best deep learning algorithms and reveal foundation model effectiveness for predicting genetic perturbations. *Bioinformatics*, 41(6):btaf317, 2025. doi: 10.1093/bioinformatics/btaf317.

[24] E. Kernfeld, Y. Yang, J. S. Weinstock, A. Battle, and P. Cahan. A comparison of computational methods for expression forecasting. *Genome Biology*, 26:388, 2025. doi: 10.1186/s13059-025-03840-y.

[25] G. Csendes, G. Sanz, K. Z. Szalay, and B. Szalai. Benchmarking foundation cell models for post-perturbation RNA-seq prediction. *BMC Genomics*, 26:393, 2025. doi: 10.1186/s12864-025-11600-2.

[26] C. Ahlmann-Eltze, W. Huber, and S. Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02772-6.

[27] A. Wenteler et al. PertEval-scFM: Benchmarking single-cell foundation models for perturbation effect prediction. In *Proceedings of the 42nd International Conference on Machine Learning (ICML), PMLR*, volume 267, 2025. URL https://proceedings.mlr.press/v267/wenteler25a.html.

[28] Z. Wei, Y. Wang, Y. Gao, et al. Benchmarking algorithms for generalizable single-cell perturbation response prediction. *Nature Methods*, 23(2):451–464, 2026. doi: 10.1038/s41592-025-02980-0.

[29] R. Viñas Torné, M. Wiatrak, Z. Piran, others, and M. Brbić. Systema: a framework for evaluating genetic perturbation response prediction beyond systematic variation. *Nature Biotechnology*, 44:1050–1059, 2026. doi: 10.1038/s41587-025-02777-8.

[30] Y. Roohani, K. Huang, and J. Leskovec. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology*, 42:927–935, 2024. doi: 10.1038/s41587-023-01905-6.

[31] I. Kendiukhov. Systematic Evaluation of Single-Cell Foundation Model Interpretability Reveals Attention Captures Co-Expression Rather Than Unique Regulatory Signal, 2026. arXiv:2602.17532.

[32] A. DenAdel, others, and L. Crawford. Evaluating the role of pretraining dataset size and diversity on single-cell foundation model performance. *Nature Methods*, 2026. doi: 10.1038/s41592-026-03120-y.

[33] M. Borenstein, L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. *Introduction to Meta-Analysis*. Wiley, 2009. doi: 10.1002/9780470743386.

[34] M. D. Luecken et al. Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, 19:41–50, 2022. doi: 10.1038/s41592-021-01336-8.

[35] C. Ergen, others, and N. Yosef. Consensus prediction of cell type labels in single-cell data with popV. *Nature Genetics*, 56:2731–2738, 2024. doi: 10.1038/s41588-024-01993-3.

[36] S. Cultrera di Montesano et al. Improving atlas-scale single-cell annotation models with hierarchical cross-entropy loss. *Nature Computational Science*, 6:243–249, 2026. doi: 10.1038/s43588-025-00945-z.

[37] C. Yi et al. Benchmarking deep learning methods for biologically conserved single-cell integration. *Genome Biology*, 26:398, 2025. doi: 10.1186/s13059-025-03869-z.

[38] T. Stuart, A. Butler, P. Hoffman, C. Hafemeister, E. Papalexi, W. M. Mauck III, Y. Hao, M. Stoeckius, P. Smibert, and R. Satija. Comprehensive integration of single-cell data. *Cell*, 177(7):1888–1902, 2019. doi: 10.1016/j.cell.2019.05.031.

[39] X. Hu et al. Reference-informed evaluation of batch correction for single-cell omics data with overcorrection awareness. *Communications Biology*, 8:521, 2025. doi: 10.1038/s42003-025-07947-7.

[40] V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi: 10.1007/b106715.

[41] A. N. Angelopoulos and S. Bates. Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/2200000101.

[42] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *ICML*, 2017. doi: 10.48550/arXiv.1706.04599. arXiv:1706.04599.

[43] M. López-De-Castro, A. García-Galindo, J. González-Gomariz, and R. Armañanzas. Conformal inference for reliable single-cell RNA-seq annotation. *Bioinformatics*, 41(10):btaf521, 2025. doi: 10.1093/bioinformatics/btaf521.

[44] L. Theunissen, T. Mortier, Y. Saeys, and W. Waegeman. Uncertainty-aware single-cell annotation with a hierarchical reject option. *Bioinformatics*, 40(3):btae128, 2024. doi: 10.1093/bioinformatics/btae128.

[45] H. Boström. crepes: conformal regressors and predictive systems, 2024. URL https://github.com/henrikbostrom/crepes. Python package.

[46] MAPIE contributors. MAPIE: model-agnostic prediction interval estimator, 2024. URL https://github.com/scikit-learn-contrib/MAPIE. Python package, v1.4.

[47] Yuxuan Wang, Yuhan Chen, Luqi Yang, Cheng Wang, Jinpu Cai, and Hongyi Xin. Benchmarking niche identification via domain segmentation for spatial transcriptomics data. *bioRxiv*, 2026. doi: 10.64898/2026.02.27.708202. https://github.com/WYXNICK/spatial-niche-benchmark.

[48] C. Wang, H. Cui, A. Zhang, R. Xie, H. Goodarzi, and B. Wang. scGPT-spatial: Continual pretraining of single-cell foundation model for spatial transcriptomics. *bioRxiv*, 2025. doi: 10.1101/2025.02.05.636714.

[49] Q. Blampey, H. Benkirane, N. Bercovici, K. Mulder, G. Gessain, F. Ginhoux, F. André, and P.-H. Courtède. Novae: a graph-based foundation model for spatial transcriptomics data. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02899-6.

[50] S. Morabito et al. Single-nucleus chromatin accessibility and transcriptomic characterization of Alzheimer's disease. *Nature Genetics*, 53:1143–1155, 2021. doi: 10.1038/s41588-021-00894-z.

[51] T. M. Norman, others, and J. S. Weissman. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365:786–793, 2019. doi: 10.1126/science.aax4438.

[52] S. Kommu et al. Prediction of gene regulatory connections with joint single-cell foundation models and graph-based learning. *Bioinformatics*, 41:i619–i627, 2025. doi: 10.1093/bioinformatics/btaf217.

[53] Y. Luo et al. Benchmarking RNA velocity methods across 17 independent studies. *Cell Reports Methods*, 2026. doi: 10.1016/j.crmeth.2026.101367.